



Affirmed Networks Overview Guide

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- Features
- Acuitas Service Management
- Virtualized Hardware Environment
- Software Architecture
- Compliance

Affirmed Networks Inc.

35 Nagog Park
Acton, MA 01720
USA

Tel: +1-978-268-0800

www.affirmednetworks.com

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Preface

This guide provides an overview of the Affirmed Networks Mobile Content Cloud™ (MCC), Acuitas Service Management System™, and the Command Line Interface (CLI) used to configure, control, and monitor services in the network. This guide also includes information on the virtual hardware and software architecture and supported features.

***Note:** The Affirmed Networks Mobile Content Cloud™ (MCC) is the mobile services software solution that operates on virtual platforms such as blade systems.*

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide), installation staff, and other qualified service personnel who want to learn about the Affirmed Networks solution and associated features and functions. The audience should also be familiar with Long Term Evolution (LTE), and 3G or 4G networks. It may also be helpful to have a basic understanding of the HTTP protocol, TCP/IP stack, and the use of MMS services.

Release Information

See the release notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Use This Guide

This guide contains the following chapters and appendices:

Chapter/Appendix	Describes
Chapter 1, Mobile Content Cloud Overview	An overview of the Mobile Content Cloud solution and supported features and services.
Chapter 2, Acuitas Service Management System	An overview of the Acuitas Service Management System features and functions such as applications, modules, statistics, authentication, authorization, and accounting (AAA), and online help.
Chapter 3, MCC Virtualized Hardware Environment	An overview of the virtualized hardware environment and supported VM hardware configurations.
Chapter 4, MCC Software Architecture	An overview of the virtualized software architecture including dynamic scaling considerations and redundancy.
Appendix A, Statement of Compliance	The MCC Statement of Compliance (SoC) with Technical Specifications (TS), standards, and Request for Comments (RFC).

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menus	Times New Roman bold font	Use configure_ha to configure... Enter y . To continue, press Enter .
Selecting a menu item	Menu => Option	Select Security => Group Management .
Variable parameters for which you supply values	<i><courier bold italic blue font></i>	configure_dr convert slave --masterhost <host IP>
User input	Courier bold blue font	cd /opt/Affirmed/NMS/bin
Screen messages, system output, sample source code	Courier regular blue font	Checking for License...
Required keywords and arguments in square brackets	[]	cluster [id] node [id] reboot
Optional keywords and arguments in curly brackets	{ }	{cdr-file-name}
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.	 	Is this correct? [y n]
Keywords representing a single element in angle brackets	< >	<host name>
Asterisk after a CLI object indicates a required object.	*	name*
Variables, book and chapter titles, file path names, new terms, and emphasized text	<i>Times New Roman italic font</i>	See the <i>Acuitas User's Guide</i> .

Tip: Helpful suggestions.

Note: Additional information that may apply to the subject text.



CAUTION: Proceed carefully to avoid possible equipment damage or data loss.

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See the *Affirmed Networks Documentation Library* for a complete list of Affirmed Networks product documentation.

MCC Software Versioning

Affirmed Networks MCC software versions support the following numbering scheme:

<Major Version>.<Minor Version>.<Maintenance Number>.<Patch Number>-<Build Number>, for example: 7.1.1.2-10

Version	Description
<Major> Example: 7.0	Identifies a major release version number and increments when the release contains major features.
<Minor> Example: 7.1	Identifies a minor release version number and increments when the release contains minor features.
<Maintenance> Example: 7.1.1	Identifies a maintenance release number and increments when the release contains small fixes as part of continuous software improvement.
<Patch> Example: 7.1.1.2	Identifies a patch release number and increments when a release contains hot fixes.

Version	Description
<Build-number> Example: x.x.x.x-10	Identifies a build number and increments every time a build is released.

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Mobile Content Cloud Overview

This section provides an overview of the Affirmed Networks Mobile Content Cloud™ portfolio of products and services used to configure, control, and monitor services in the network.

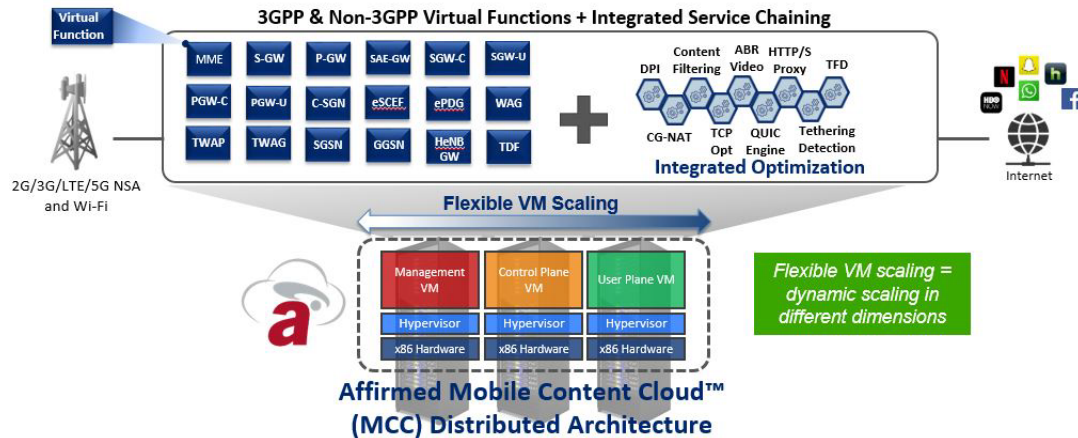
Mobile Content Cloud Solution

The Affirmed Networks Mobile Content Cloud (referred to as Mobile Content Cloud or MCC) solution provides Affirmed Networks services in a virtualized environment, with software executed on multiple virtual machines that are separate from the underlying hardware resources. Network virtualization (network functions implemented as virtualized software applications in an Operator's data center) simplifies network operations and reduces capex.

The MCC solution is targeted toward Mobile Network Operators (MNO), Mobile Virtual Network Enablers (MVNE), Mobile Virtual Network Operators (MVNO) and Machine-to-Machine (M2M) Operators.

The MCC is a virtualized EPC (vEPC) + multi-VAS solution developed to simplify and rationalize the mobile packet core and value-added services networks. The MCC consists of a cluster of differentiated Virtual Machines (VMs) that collectively provide a single composite-VNF (C-VNF). The composite-VNF functions as a single managed Network Element (NE) and can simultaneously function as any one or more VNFs provided by Affirmed Networks. See [Affirmed Networks MCC Solutions Portfolio](#).

[Figure 1-1](#) shows the vEPC architecture.

Figure 1-1 Affirmed Networks vEPC Architecture

The MCC delivers the complete Affirmed Networks solution on industry-standard servers residing in a mobile Operator's data center. Content functions of the MCC solution can be flexibly chained using the Affirmed Open Workflow and integrated with existing network elements using standard 3GPP interfaces to offer a variety of services including content services, monetization, and enhanced management.

The MCC provides full virtualization of both the functional and hardware layer. The functional layer collapses all of the gateway and Value-Added Services (VAS) functions into one software solution. The hardware layer consolidates the mobile services core network on commercial off the shelf IT (COTS) servers or blade-centers from well-known IT vendors such as HP, IBM, and Dell.

The MCC runs on virtually any Intel x86 server or blade centers using Intel Xeon multi-core processors such as HP DL380 or HP c7000 servers and IBM PureFlex blade-center systems. MCC is available on hypervisors such as VMware ESXi, KVM, and OpenStack. MCC supports standard vSwitch deployments and high-performance direct Single Root I/O Virtualization (SR-IOV) deployments. MCC can also co-exist with other third-party VNFs on the same NFVI. See [Chapter 3, MCC Virtualized Hardware Environment](#) for more information.

The Affirmed Networks VNF solutions integrate into network environments using standard 3GPP and IETF interfaces and can be deployed in different configurations and business models such as:

- Dedicated servers/blade-centers
- Operator data centers (with other VNFs)
- Hosted in Telco-Clouds (VNIAaaS, VNPaaS, VNFaaS)
- Co-hosted with third-party VNFs

The MCC architecture supports multiple Virtual Routing and Forwarding (VRF) instances enabling Operators to define multiple types of services as part of each VRF context.

MCC supports Dynamic Capacity Scaling. Operators can configure KPIs (number of sessions or subscribers, CPU load, throughput, and so on) in the system which triggers the MCC to add or remove VMs automatically in the Operator's data center or telco cloud, based on real-time traffic evolution.

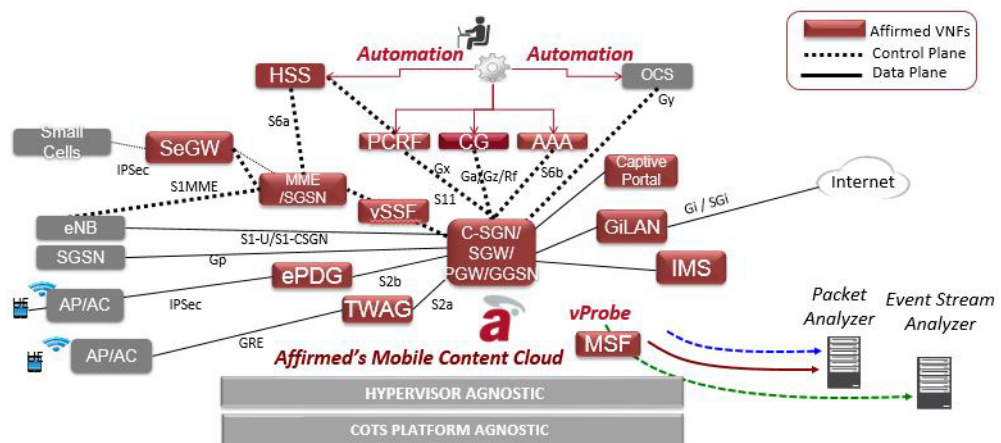
The main functions provided by the MCC solution are mobile gateway and value-added services with subscriber workflow services chaining. Content management in the system includes the basic mobile CDN functions of caching. The system also supports additional CDN functions that enable Operators to provide revenue-generating services. See the *Affirmed Networks Gateway Administration Guide* for more information.

Figure 1-2 shows the relevant 3GPP interfaces and non-3GPP access networks supported by MCC. These interfaces provide the following services and support:

- 2G, 3G, 3G+, and 4G mobile data services to home and roaming subscribers
- Voice support in the LTE radio network such as IMS-based VoLTE
- Seamless handoff and voice traffic offload to the trusted and untrusted Wi-Fi access network for consistent voice support and handovers between LTE and Wi-Fi and vice versa

For information about the interfaces supported in MCC CUPs, see [Control and User Plane Separation \(CUPS\)](#).

Figure 1-2 MCC Solution - Supported Interfaces



Supported Services and Features

The Affirmed Networks solution supports the following services and features:

Table 1-1 Affirmed Services and Features

Service/Function	Description
3GPP Mobile Gateway Services	■ SGW ■ PGW ■ GGSN ■ SGSN ■ GiGW ■ C-SGN ■ Primary and secondary PDP context (GGSN) ■ PGW-C/U, SGW-C/U, and collocated SAEGW-C/U (see Control and User Plane Separation (CUPS) Support)
Non-3GPP Access Gateway Services	■ TWAG (trusted WLAN access) ■ ePDG (untrusted Wi-Fi access)
Lawful Intercept/Subscriber Tracing	■ X1_1, X2, and X3 interfaces ■ Support for persistent and non-persistent multiple X2/X3 delivery destinations ■ Support for provisioning a range of TCP source ports for each CALEA X2 and X3 interface ■ Lawful Intercept support on gateway network elements ■ CUPS Lawful-Intercept support for dynamic LI targets on User Plane Function (UPF) ■ CUPS Lawful-Intercept support for Packet Data Summary Report records
Emergency services over WLAN	■ Support on ePDG for R14 emergency services calls

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
Interfaces	■ Gi, SGi, Gn, Gp, Gx, Gz, Gy, VF DCCA, Gz (Diameter) ■ S1-U, S4, S10, S11, S4, S12, S5/S8, S6b, S2a, S2b, Sd, STa, SWm, SWn, SWu ■ Sxa, Sxb (see Control and User Plane Separation (CUPS) Support) ■ Bp ■ RADIUS ■ Lawful Intercept (X1, X2, X3) ■ Geographic Redundancy support ■ Support for customizing the Access Point Name (APN) reported on external interfaces, such as Gx, Gy, Gz, S6b, RADIUS, Ga (GTPP) and Local ■ Support for changing the default AVP flag values for Diameter AVPs on a specific Diameter interface (Gx, Gy, Gz, SWm, STa, and S6b) ■ Support for using Stream Control Transmission Protocol (SCTP) as a transport protocol on the Gx, Gy, Gz/Rf, S6b, and SWm/STa Diameter interfaces ■ Support on SWm interface for multiple primary active and secondary peer pairs ■ Support for multiple primary active and secondary peer pairs on the S6b interface. Operators can configure up to four pairs of peers (four primary peers and four secondary peers) on the S6b interface within a Diameter server group. ■ Support for IKEv2 Mobility and Multihoming Protocol (MOBIKE) on the SWu interface (ePDG) as defined in RFC 4555 ■ Gx interface support for Application Detection and Control (ADC) ■ Gx and Gy interface support for CUPS ■ Gx and Gy interface enhancements to support 5G NSA ■ Gx interface support for content-filtering on GiGW as indicated by PCRF ■ Gx interface rejoin enhancement for notifying the PCRF when an OCS failure occurs ■ Support for multiple PDN connections for the same user (same IMSI) with the virtual Slice Selection Function (vSSF) on the S5/S8 interface for co-located SGW/PGW (SAEGW)

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
Infrastructure/Protocols	■ IPv4, IPv6 ■ IPv6 SFTP file transfer ■ System log-file streaming for IPv6 addresses ■ UE IP address allocation using local IP address pools ■ UE IP address allocation using AAA (RADIUS) and DHCP ■ MCM redundancy (1:1) with enhanced failover operations on primary/secondary ■ Fault isolation, Session cleanup ■ Fault management ■ Alarm throttling ■ Protocol logging ■ Event-log filter field includes support for macaddress and NAI keywords as (subscriber) logging filters ■ Configuration and operational measurements, logs, IPv4, IPv6, dual stack IPv4/IPv6 PDP, and EPS bearers ■ Peer overload control ■ GTPv0, GTPv1, and GTPv2 ■ Enhanced SGW/PGW/GGSN and GTP Proxy GTP-C path management ■ Reliability – Geo-resiliency – Session retention – Overload control – N-way redundancy – Online software upgrade ■ Geo-redundancy support for NAT port chunks in Guaranteed mode of operation ■ Failover command for geographic-redundancy groups using hot-auto mode ■ Secondary loopback IP address for the local Diameter node and Diameter peer to support SCTP Multi-homing (and SCTP transport type). The primary and secondary loopback IP addresses can be IPv4 or IPv6. ■ Support for a separate network-context loopback IP for export users to access and transfer export data using SCP/SFTP via a north-south interface rather than through the standard management-context, mgmt-ip-address.

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
Authentication	■ Support for IPSec IKEv2 message fragmentation (RFC 7383) which allows IKEv2 to interoperate with IKEv2 peers that support IKE fragmentation ■ Trusted non-3GPP access uses Extensible Authentication Protocol (EAP) for secure authentication and 802.11 for airlink security ■ Untrusted non-3GPP access: – Internet Key Exchange (IKE) using the IPSec/IKEv2-based SWu interface between the ePDG and WLAN UE – Extensible Authentication Protocol - Authentication and Key Agreement (EAP-AKA) between the 3GPP AAA server and the WLAN UEs – Layer 2 Tunneling Protocol (L2TP) through Extensible Authentication Protocol - Generic Token Card (EAP-GTC) for virtual private networks (VPNs) or ISP services that use L2TP tunnels – Public Key Infrastructure (PKI) authentication used for the exchange of digital certificates between the UE and ePDG, including local, ca-trust, and intermediate-trust certificate types ■ Support for IKEv2 Mobility and Multihoming Protocol (MOBIKE) on the SWu interface (ePDG) as defined in RFC 4555 ■ Support for MAC algorithms at the maximum pki ssh security level ■ Support for the RC4MD5 encryption standard in HHE request/response headers ■ Terminal Access Controller Access Control System Plus (TACACS+) for remote authentication, authorization, and accounting for user management and access to the MCC ■ Support for Secure Neighbor Discovery protocol (SEND) as described in RFC 3971/3972 for secure Neighbor Discovery ■ Acuitas AAA; see Acuitas Service Manager and VNF Manager

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
RADIUS Gateway Services	■ RADIUS peer overload control, which limits the number of transactions transmitted per time interval from the gateway to the RADIUS server ■ Support for mirroring RADIUS Accounting-Request messages, sent from a RADIUS client to the GiGW ■ Accounting (RADIUS) - Data profile selection for GiGW session using RADIUS vendor-specific attribute ■ RADIUS-based authentication server for subscriber authentication and subscriber profile retrieval ■ Subscriber Aware GiGW support for non-standard RADIUS Health Checks ■ Support for configuring RADIUS peers with Virtual Routing and Forwarding (VRF) lightweight network contexts.
Networking	■ Support for multiple IP interfaces (IPv4/IPv6) sharing the same IP subnet within a single network-context ■ IPv4-IPv4 NAT ■ Enhanced NAT statistics ■ Flow EDRs include the NAT IP when NAT translation is performed ■ Fixed/deterministic NAT for PGW, GGSN, and GiGW (subscriber-aware mode) ■ Full Cone support for NAT functionality supporting Endpoint Independent Mapping (EIM) and NAT-filtering schemes ■ CGNAT status and statistics reporting ■ CGNAT flow-level profile settings for network-context nat-pool level and subscriber-firewall-flow level ■ Support for ICMP and PPTP on subscriber-firewall-flow alg object ■ Global firewall enhancements ■ Support for Subscriber Aware GI Gateway (SA-GiGW) to interwork with Service Function Chaining Proxy (SFCP) ■ Multiple VRF contexts for Gn, Gi, Management (EMS), AAA, PCRF, charging servers, and other interfaces ■ Support for 2K VRFs or BGP VPN instances (VRF-Lite Network-Context)

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
Networking (<i>continued</i>)	■ Application Layer Gateways (ALGs) for FTP, TFTP, SIP, and RTSP ■ FTP Application Layer Gateway (ALG) support for NAT44 and NAT64 ■ IPSec tunneling support for NAT44 and NAT66 ■ Session Initiated Protocol (SIP) Application Layer Gateway (ALG) support for NAT44 ■ Support for ALGs without NAT ■ Support for NAT 66 ■ Support for 1:1 NAT or static NAT in Guaranteed, Deterministic, and Statistical NAT pools with CUPS User Plane Function (UPF) ■ Traffic Anchoring Mode (TAM) on SSM/ISM ■ TAM with Multi-hop Border Gateway Protocol (MH-BGP) ■ Multi-Active routing for BGP Multi-hop Peering ■ Support for distributing BGP peers to the administratively enabled CCM/CSMs as part of multi-active routing HA ■ Support for multiple IP interfaces and BGP sessions in the shared subnet (SH-BGP) ■ Support for IPv4 and IPv6 for SH-BGP within a single network-context ■ Support for multi-active mode with single hop shared-subnet configuration. (Supports multiple BGP peering to the same remote address for SH-BGP with shared subnets.) ■ Dynamic routing (OSPFv2) ■ OSPFv2 (IPv4) and OSPFv3 (IPv6) ■ OSPF route summarization ■ MPLS (GTP-U and GTP-C) ■ L2TP, GRE, BGP, BFD, IPSec ■ IPSec and egress fragmentation

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
Online and Offline Charging	■ Time, volume, time and volume ■ Basic and SDF-based charging ■ Support for activating/deactivating metering services for rating groups and service rules associated with a Charging-Rule-Base Name (CRBN) profile ■ Support for selecting an online-charging interface with a CRBN profile ■ Support for moving predefined service rules across predefined rating groups ■ Support for including/excluding the service ID in service data containers (SDC) when generating CDRs ■ CDR generation for offline charging ■ Enhanced CDR file-management profile parameters ■ SGW CDR generation based on a list of configured IMSI ranges ■ Customizable CDRs and Gy interface via templates ■ Configuration option to treat CDR Data Record Transfer Response as an Echo Response over Ga interface ■ Configuration option to specify a 3GPP 32.298 version used to format ASN.1 local CDR fields ■ CDR mediation for local ASN.1 and GTPP CDRs ■ PGW Gy usage reporting and ASN.1 Local PGW CDR enhancements ■ Cumulative statistics for data usage at the interface, service rule, and bearer level ■ Gy, DCCA ■ RADIUS ■ GTPP ■ L7 (HTTP) based QoS ■ L7 (HTTP) based charging ■ Event/transaction-based charging ■ Offline charging support for CUPS (local interface, Ga interface, and Gz/Rf interface) ■ Support for overriding certain offline-charging interface parameters using Charging-Rule-Base-Name (CRBN) from PCRF ■ Support for deferring establishment of online-charging session and request for quota until after first SDF received. ■ Support for PCRF dynamic allocation of a rating group ■ Support for overriding the charging-characteristics (CC) value received in the Create Session Request message for a subscriber based on the subscriber type.

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
Workflow	■ SPI and DPI signature analysis ■ Subscriber firewall (URL list based) ■ Access Control List (ACL) ■ Enhanced tethering detection logic using TCP-IP fingerprinting and GQUIC user agent fingerprinting ■ Support for controlling tethering for an individual APN based on the presence of configurable strings in the acw entitlement vendor-specific attribute (VSA) received from the PCRF ■ DNS protocol analyzer ■ Subscriber anonymization/aliasing ■ APN mapping and aliasing ■ APN mapping/redirection using RADIUS ■ Workflow evaluation and tracing ■ Time of day support for workflow services ■ Subscriber analyzer remapping based on UE and network triggers ■ Dynamic and static service chaining
Deep Packet Inspection	■ Packet classification for WAP 1.0, 2.0, and MMS ■ SPI ■ DPI ■ DPI classification model enhancements with protocol-application-rule-group configuration ■ Support for Deep Packet Inspection (DPI) in multi-protocol-proxy policies ■ DPI detection for new applications, protocols, and sub-protocols for Application Rule Group and Deep Packet Inspection ■ DPI detection of the IMO application-related protocol traffic to classify subprotocols (audio/video) traffic separately from rest of IMO traffic ■ DPI application statistics on flow and links (bytes and packets) usage on Google sub-protocols ■ Heuristic signature analysis ■ Seamless signature upgrades

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
Traffic Management	■ Uplink and downlink bearer rate enforcement ■ Dynamic policy configuration and enforcement ■ Rate-limit bandwidth based on a configured subscriber group ■ Acuitas support for configuring Rate Limiting Subscriber Group ■ Mapping of ePDG QoS Class Identifiers (QCIs) to specific Differentiated Services (DiffServ) Code Point (DSCP) and IEEE 802.1Q (DOT1Q or 1Q) values between the SWu and S2b interfaces ■ VoLTE bearer setup with dual RAR enabling the PGW to handle back-to-back RAR messages from the PCRF on the Gx interface during a dedicated bearer setup ■ Extended Quality of Service class identifier (QCI) range from 1-255
Handoffs/Handovers	■ Inter eNodeB handoff ■ X2 and S1 handoff ■ Inter SGW handoff ■ Inter SGSN handoff ■ 2G-4G (inter SGSN and eNodeB) handoff ■ 2G-3G (inter SGSN and eNodeB) handoff ■ 3G-4G (inter SGSN and eNodeB) handoff ■ Wi-Fi to LTE and LTE to Wi-Fi handovers between the untrusted non-3GPP IP access network and the 3GPP LTE network via S2b interface ■ Delay timer for LTE to Wi-Fi (non-3GPP) handovers ■ Seamless authentication and handoffs between trusted WLAN access networks and the EPC via the GTPv2-based S2a interface ■ Support for inter-system handover for a QoS PGW-triggered QoS upgrade, increasing the APN AMBR when a session transitions from 3G to 4G ■ Support for Inter Radio Access Technologies (IRAT) Handover (SGW→SGSN or SGSN→SGW) when the PGW receives a Modify Bearer Request from the SGSN with the same TEID and IP Address as the SGW and a GTP version change occurs, for example, GTPv1 to GTPv2

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
General Multi-Protocol-Proxy Features	■ IP Transparency ■ Traffic steering ■ Proxy steering group enhancement allows peers with the same IP address to be reachable through different network-contexts ■ DPI support in multi-protocol-proxy policies ■ Graceful restart of proxy faults and process redundancy ■ HTTP Transaction Data Records (HTDR) ■ HTTP Event Data Records (EDR) including information about the content detected in the HTTP response ■ TCP optimization – TCP congestion-triggered optimization enhancements – TCP performance statistics – Configurable TCP initial receive window and initial congestion window and increased maximum segment size – Early Retransmit (ER) and Tail Loss Probe (TLP) support for TCP optimization during short Web transactions ■ Support for user-space-networking (TCP and UDP) integration for all HTTP/HTTPS/TCP/UDP proxy functionality for IP transparent and non-IP transparent proxy connections (ISM only) ■ Enhancements to services multi-protocol-proxy toll-free-profile

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
HTTP-Proxy	■ Transparent and Explicit HTTP-proxy support ■ HTTP 1.0 and 1.1 clients ■ HTTP 1.0 and 1.1 connections toward the origin server ■ Persistent and Pipelined HTTP client connections ■ IPv4 and IPv6 support ■ Chunked transfer format from clients ■ HTTP range requests ■ Extensive configuration and statistics support ■ HTTP Rule Group - Request and Response rules ■ Gzip compression ■ URL rewriting ■ HTTP header enrichment ■ Long Term Mobile ID (LTMI interface support) ■ Adaptive HTTP video streaming ■ Image Optimization (.jpg, .gif, .png) ■ Minification of HTML, CSS and JavaScript – Video pacing – MP4, FLV and WebM container support ■ DNS preemption ■ Fraud prevention ■ Support configuration to truncate URL size in EDRs ■ HTTP Proxy statistics listing HTTP responses for each detected media type
HTTPS/TCP Splicing	■ Support for estimating optimization byte savings attributed to rate-limited HTTPS video traffic and displaying the savings in logs, statistics, and HTDRs ■ TCP proxy intelligent-buffering flow control to limit the traffic differential between the Radio network and the Core/Internet ■ SD/HD video detection in HTTPS ■ Support for Application Layer Protocol Negotiation (ALPN)-based methods of identifying HTTPS content ■ ABR shaping for multi-stream HTTPS connections to the same server enabling Operators to apply rate limiting (shaping) to all flows concurrently, as a group, for a single logical video

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
QUIC/UDP-Proxy	■ Support for QUIC video optimization – QUIC handshake decoding – QUIC domain-name based service rules – QUIC traffic shaping (rate-limiting) – QUIC data records ■ QUIC video savings estimation ■ Ability to decode QUIC protocol handshake which uses Google version 44 protocol format and supports Google QUIC Hello statistics in udp-proxy quic protocol
DNS-Proxy	■ DNS caching
Content-Filtering: HTTP Content Filtering	■ Category-based content filtering ■ Malware and phishing protection ■ Content filtering support for QUIC SNI and/or server certificate information, which provides the ability to block/allow QUIC traffic based on the configured filter service-rules ■ URL enrichment via 307 direct ■ Global blacklist URLs applicable to all subscribers ■ Content filter service workflow configuration ■ Configuration to block, allow, or redirect HTTP content ■ HTTP redirection based on the HTTP request method within the MCC ■ HTTP redirection for dynamic or externally activated predefined rules based on information received from the PCRF in the Gx-TMO-Redirect-Server AVP ■ Subscriber-specific content filtering ■ DPI based content filtering for subscriber workflows ■ HOST and URI filtering ■ Safe-Internet URL blocking ■ Google safe-search call flow ■ Exact match and regex in global blacklist URLs ■ Up to 32k URLs in the global blacklist ■ Dynamic parental control ■ Content type filters

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
Content-Filtering: HTTP Content Filtering (cont'd.)	■ Symantec related functionality: – automatic downloads of URL, IWF, malware and phishing databases – automatic HTTP request categorization to approximately 100 categories – online category blocking and redirection filters – real-time categorization for responses with dynamic content – multi-language support – Internet Watch Foundation filtering – malware filtering – phishing filtering ■ Content-categorization filtering for HTTPS traffic ■ Content-categorization filtering for QUIC traffic
Content-Cache	■ Strict cache primitives per protocol ■ In-memory and on-disk cache ■ Cache expiration for web cache ■ Fine-grained configuration for web caching ■ Cache revalidation of popular objects ■ LFU and LRU-based cache eviction ■ Selective purging of cache objects ■ Cache reloading across reboots ■ Graceful handling of cache objects during process and system faults ■ Write avoidance algorithm ■ Graceful handling of disk failures ■ Cache override ■ Popularity metrics based caching (object size, hit rate) ■ Capability to bypass cache ■ Forced cache access notification HTTP Video Caching ■ HTTP progressive download caching ■ Extensive configuration and statistics ■ Graceful handling of disk failures ■ Cache override ■ Popularity metrics based caching (object size, hit rate) ■ Forced cache access notification

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
HTTP-Server	■ Entitlement server – Simple entitlement server – Secure entitlement server ■ Secure HTTP Identity Protocol (SHIP) enhancements support multiple applications offered by a single content provider by differentiating URLs with the same domain name. The applications (inferred from different URL paths) offered by a single content provider (inferred from domain name) are treated as different content providers.
MMS and WAP Services	■ MMS proxy and transcoding ■ WAP2.0 gateway ■ MM1 proxy ■ MMS transcoding ■ STI/MM11 interface ■ MSISDN anonymization ■ WAP Gateway
Enterprise Tunnels	■ GRE; includes keepalive support on heavyweight tunnels that use the GRE tunnel state updated by GRE keepalives. ■ MPLS ■ L2TP ■ IPSec; includes support for dynamic tunnel management and redundant tunnel groups ■ Subscriber EDRs

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
Acuitas Service Manager and VNF Manager	■ FCAPs support for all services including gateways, value added services, content filtering, charging, and security ■ Acuitas Security Module AAA support: – TACACS+ – LDAP – RADIUS ■ Topology ■ Performance dashboard ■ Performance collection, reports generation and dashboard configuration ■ Network inventory management ■ Security, role-based views for configuration ■ Log upload ■ EMS/NE software upgrade and rollback ■ Service workflow configuration ■ Context sensitive help ■ Device configuration and backup and restore ■ EMS backup and restore ■ High availability ■ Disaster recovery and geographic redundancy ■ Fault management ■ Performance management ■ Configurable Performance Manager thresholds for SSF counters ■ Support for HTTP Filter File Management to create, modify, delete URL list element and copy to the MCC ■ Support for CGNAT error statistics

Table 1-1 Affirmed Services and Features (Continued)

Service/Function	Description
Control and User Plane Separation (CUPS) Support	■ Separate control plane functions (PGW-C, SGW-C) and user plane functions (PGW-U, SGW-U) and collocated SAEGW-C/U control and user plane functions ■ Sxa interface between SGW-C and SGW-U ■ Sxbc interface between PGW-C and PGW-U (the Sxc interface functions as part of the Sxb interface) ■ Sxabc interface between collocated SAEGW-C and SAEGW-U ■ Packet Forwarding Control Protocol (PFCP) ■ PFCP session monitoring on Sx interfaces ■ PFCP protocol logging ■ Support for Sx node-related procedures: Heartbeat messaging and Sx Association procedures ■ User Plane Function (UPF) selection based on existing subscriber analyzer keys and additional UE capabilities key ■ Geo-Redundancy support on Control Plane Function (CPF) ■ Lawful-Intercept support for dynamic LI targets on User Plane Function (UPF) ■ Lawful-Intercept support for Packet Data Summary Report records ■ CUPS support on Gx and Gy interfaces See Control and User Plane Separation (CUPS).
5G Non-Standalone New Radio Dual Connectivity	■ Support for 5G devices to connect to 5G frequencies over 4G LTE networks for improved data throughput and migration to 5G ■ Gx and Gy interface enhancements to support 5G NSA ■ 5G NSA Support of UP Selection Indication Flags IE on the S4/S11 and S5/S8 Interfaces ■ 5G NSA Support of Secondary RAT Usage Data Reporting IE on the S11 and S5/S8 Interfaces See 5G Non-Standalone New Radio (5G-NSA NR) Dual Connectivity.

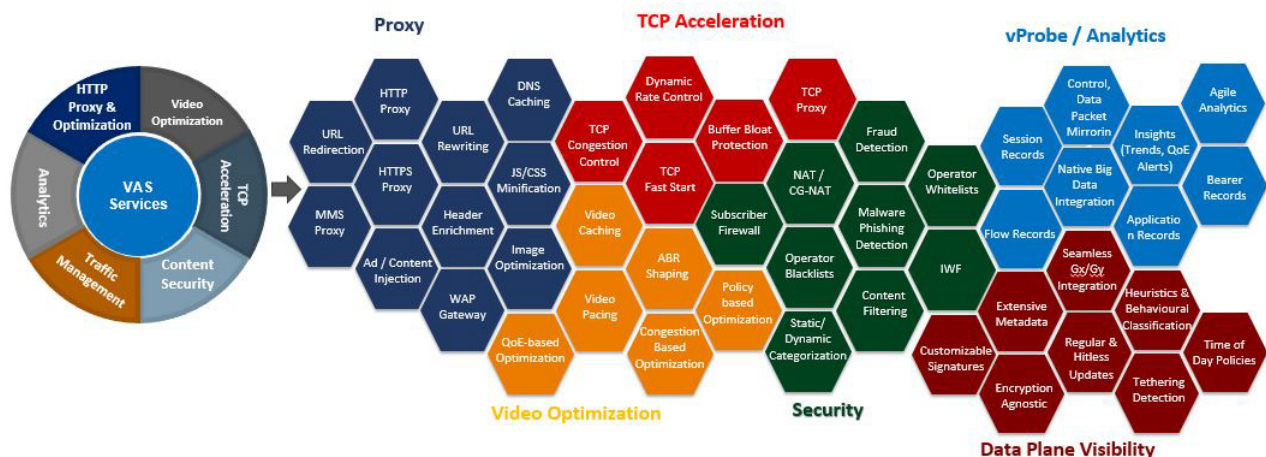
Affirmed Networks MCC Solutions Portfolio

The section describes the following VNFs and services supported in the MCC portfolio:

- Serving Gateway
- Packet Data Network Gateway
- Gateway GPRS Support Node
- GiLAN-Gateway (GiGW)
- Serving GPRS Support Node
- C-SGN
- TWAG and TWAP
- ePDG
- Control and User Plane Separation (CUPS)
- 5G Non-Standalone New Radio (5G-NSA NR) Dual Connectivity
- Acuitas Service Management System

The MCC VNFs include access termination capabilities as well as value-added network functions (shown in [Figure 1-3](#)) such as NAPT, Firewall, multi-protocol proxy, content caching, video optimization, TCP optimization, minification, MMS proxy, SMTP proxy, and so on. This solution provides the flexibility to operate as a standalone entity supporting one of these functions or as an integrated solution supporting many functions.

Figure 1-3 MCC Value Added Services



Serving Gateway

The Affirmed Networks Serving Gateway (SGW) is the point of interconnect between the radio-side and the EPC. This gateway serves the UE by routing incoming and outgoing IP packets. It is logically connected to the PGW.

The SGW acts as the mobility anchor for the user plane during inter-eNodeB handovers and as the anchor for mobility between LTE and other 3GPP technologies (terminating S4 interface and relaying the traffic between 2G/3G systems and PGW). For idle state UEs, the SGW terminates the downlink data path and triggers paging when downlink data arrives for the UE. It manages and stores UE contexts, for example parameters of the IP bearer service, and network internal routing information. The SGW also performs replication of the user traffic in case of lawful interception.

Packet Data Network Gateway

The Affirmed Networks Packet Data Network Gateway (PGW) is the point of interconnect between the Evolved Packet Core (EPC) and the external IP networks. The PGW routes packets to and from the Packet Data Network (PDN). The PGW also performs various functions such as IP address / IP prefix allocation or policy control and charging. In a mobile network, a major function of the PGW is to allocate IP addresses to the user equipment during default bearer setup.

In the Long Term Evolution (LTE) architecture for the Evolved Packet Core (EPC), the PGW acts as an anchor for user plane mobility. User traffic can be filtered at the PGW for quality-of-service (QoS) differentiation among multiple packet flows. The PGW also collects charging information and forwards Charging Data Records (CDRs) for processing.

The PGW provides connectivity from the UE to the external PDN by acting as the point of exit and entry of traffic for the UE. A UE may have simultaneous connectivity with more than one PGW for accessing multiple PDNs. The PGW performs policy enforcement, packet filtering for each user, charging support, lawful interception, and packet screening. Another key role of the PGW is to act as the anchor for mobility between 3GPP and non-3GPP technologies such as WLAN and 3GPP2 (CDMA 1X and EvDO).

GGSN Support

When enabled, a PGW can also function as a GGSN and can be connected to an SGSN and process 2G and 3G session requests. The PGW supports internetworking with pre-release 8 SGSN, which means an SGSN can directly connect to a PGW and establish 3G sessions. The PGW supports session creation and handoff between 3G and 4G networks.

Gateway GPRS Support Node

The Affirmed Networks Gateway GPRS support node (GGSN) is a main component of the GPRS network. The GGSN is responsible for the internetworking between the GPRS network and external packet switched networks, such as the Internet and X.25 networks.

The GGSN is the anchor point that enables the mobility of the user terminal in the GPRS/UMTS networks. It maintains the routing necessary to tunnel the protocol data units (PDUs) to the SGSN that services a particular MS (mobile station).

The GGSN converts the GPRS packets coming from the SGSN into the appropriate packet data protocol (PDP) format (for example, IP or X.25) and sends them out on the corresponding PDN. In the other direction, PDP addresses of incoming data packets are converted to the GSM address of the destination user. The readdressed packets are sent to the responsible SGSN. For this purpose, the GGSN stores the current user profile and SGSN address in the location register. The GGSN is responsible for IP address assignment and is the default router for the connected user equipment (UE).

The GGSN also performs authentication and charging functions. Other functions include subscriber screening, IP pool management and address mapping, QoS, and PDP context enforcement. In an LTE scenario, the GGSN functionality moves to SAE gateway (with SGSN functionality working in MME). See [Serving GPRS Support Node](#) for more information.

GiLAN-Gateway (GiGW)

The GiLAN-Gateway (GiGW) is deployed between the Evolved Packet Core PDN Gateway (PGW/GGSN) and the external IP network. The GiGW communicates with the PDNGW/GGSN over the Gi-access interface and communicates towards external IP networks with the Gi-core interface. The GiGW provides subscriber-aware and subscriber-unaware value added services (VAS) without enabling MCC gateway services. This enables Operators to use their incumbent gateway while taking advantage of VAS. VAS includes enhanced charging, content filtering, content caching, DPI, video optimization, content proxy, and so on. The GiGW can be deployed in mobile networks, fixed-line networks, and Wi-Fi networks.

The GiGW also acts as the Traffic Detection Function (TDF), and supports the Sd interface to establish and terminate TDF sessions to the PCRF.

The MCC GiGW supports:

- Redundancy, distributed architecture, fault containment
- Multiple GiGW services simultaneously
- GiGW and GGSN/SGW/PGW services simultaneously

- Virtualization and hardware agnostic operation

Subscriber Aware Mode

In subscriber aware mode, the GiGW reuses the PDNGW (GGSN, PGW) subscriber architecture for MCC subscribers. The PDN Gateway subscriber state-machine is reused for the MCC subscriber session. This feature introduces a “session learning” phase where session profile information is learned. After the complete session profile is learned, the system triggers the GiGW session setup and the subscriber state-machine retrieves the user profile from the configured interface (LDAP/PCRF/OCS/RADIUS) and accordingly establishes a workflow session on the fastpath.

In this mode, subscriber session information is learned from the RADIUS accounting/authorization interface. Then, using session information, subscriber user profile information is retrieved (from PCRF/AAA/LDAP/OCS/Static Workflow Configuration), and the system provides all user subscribed value added services. In this mode, all supported Gi Gateway services can be enabled and provide differentiated services determined by the subscriber user profile.

The GiGW supports the following gateway session learner interfaces in subscriber-aware mode.

- radius-server – The GiGW works as a RADIUS server and learns the session through received RADIUS messages and creates a GiGW session.
- radius-proxy – The GiGW works as a proxy for RADIUS clients and servers and learns the session by receiving RADIUS request messages from the client and response messages from the server and creates a GiGW session.
- gigw-ldap-client – The GiGW uses the IP packet as the trigger to go to LDAP to learn the subscriber information and set up the GiGW session.
- gigw-ip-local – The GiGW uses the IP packet as the trigger and creates the GiGW session using the local configuration.

Subscriber Unaware Mode

The subscriber unaware GiGW is used to enable subscriber unaware VAS (video optimizations, DPI, and so on). In this mode, the Operator is not required to supply a RADIUS feed. The workflow sessions are created “on the fly” upon receiving the first data packet from a subscriber, and the same workflow services are applied to all subscribers. In this mode, subscriber session information is not learned, and all subscribers receive all offered Gi Gateway services, that is, there are no differentiated services. In this mode, you can enable Gi Gateway services such as DPI, analytics, and video optimization, however this mode does not support EDRs.

Supported Interfaces

The GiGW supports the following interfaces:

- GI-Access – Between GiGW and (PGW/GGSN/other GWs)
- GI-Core – Between external Internet network and GiGW
- GiGW-AAA – Between GiGW and AAA server for RADIUS proxy mode
- Gx – Between GiGW and PCRF (partially compliant)
- Gy – Between GiGW and OCS Server (partially compliant)
- Local CDR – GiGW and local CDR server
- Sd – Between GiGW and PCRF (simultaneous support for the Sd and the Gx interface is not supported)

Serving GPRS Support Node

The Affirmed Networks Serving GPRS Support Node (SGSN) is responsible for the delivery of data packets to and from the mobile stations within its geographical service area. The SGSN is responsible for packet routing and transfer, mobility management (attach/detach and location management), logical link management, and authentication and charging functions. The location register of the SGSN stores location information (for example, current cell, current VLR) and user profiles for example, IMSI, address(es) used in the packet data network) of all registered GPRS users.

The Serving GPRS Support Node / Mobility Management Entity (MME) is a fully Virtualized Network Function (VNF) according the ETSI NFV standards. It is composed of six Virtual Machines (VM) which can be dimensioned independently according to the traffic mix.

The Affirmed Networks SGSN/MME is a 3GPP Release 11 compliant SGSN and MME that provides mobility and session management operations in separated or combined modes for 2G, 3G, 3G+ and 4G mobile data subscribers. The SGSN/MME is fully virtualized and ETSI NFV compliant and can be deployed on dedicated servers or in the Cloud.

C-SGN

The Affirmed Networks Cellular Internet of Things (CIoT) Serving Gateway Node (C-SGN) is a fully virtualized EPC node that is optimized to support the C-IOT network. It is a combined EPC node that collocates EPS entities in the control and user planes (MME, SGW, PGW). The external C-SGN interfaces are the respective EPC entities supported by the C-SGN, such as the MME, SGW, and PGW.

Operators can deploy the C-SGN in the following configurations:

- Combined MME, SGW, and PGW
- Combined MME and SGW. The C-SGN works with an external PGW for small data communication between the CIoT devices and their corresponding application servers.

On top of the existing EPC capabilities of the C-SGN, it also provides and supports the following optimization for the CIoT devices:

- Narrow band eNodeBs. The C-SGN connects to the NB-eNodeB via the S1-lite interface.
- Control plane CIoT EPS optimization for small data transmission. The C-SGN supports small data transmission using the NAS signaling layer. This is also referred to as DoNAS (Data over Non Access Stratum).
- Security procedures for efficient small data transmission. The small data transmission is integrity protected and encrypted.
- SMS without combined attach for NB-IoT only UEs. The CIoT device can perform SMS without performing combined attach procedure.
- Non-IP data transmission via SGi tunneling. Added support for new PDN type used to deliver non-IP data.
- Attach without PDN connectivity. The UE can attach without activating a PDN session.
- Power savings. The C-SGN supports many features to allow the CIoT devices to conserve battery, for example, Power Saving Mode, Extended DRX, Paging Optimization, etc.

TWAG and TWAP

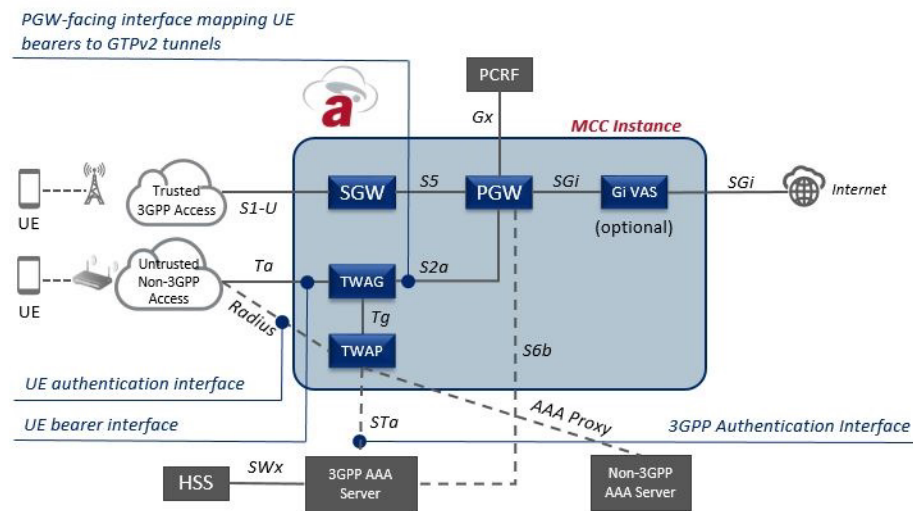
With the exponential growth of mobile data traffic over existing Radio Access Networks (RANs), offloading mobile data traffic from existing RANs to a cost-effective access network, such as Wi-Fi, is emerging as a way to manage network capacity and network load.

The MCC Trusted WLAN Access Gateway and Trusted WLAN AAA Proxy offers mobile Operators a Wi-Fi offload solution by providing user equipment (UE) authentication and subscriber management services to trusted non-3GPP attached subscribers. Trusted non-3GPP access generally refers to Wi-Fi networks controlled by mobile operators with the capability to interact directly with the EPC. Trusted non-3GPP access uses Extensible Authentication Protocol (EAP) for secure authentication and 802.11 for airlink security.

The MCC TWAG/TWAP interacts with Wi-Fi access points (APs) to identify, authenticate, and provide trusted non-3GPP subscribers access to the Internet, either directly using Non-Seamless WLAN Offload (NSWO) or through the EPC using 3GPP S2a Mobility based on GTP (SaMOG) services as defined in 3GPP TR 23.852.

Figure 1-4 shows the position and role of the network components in an MCC TWAG/TWAP deployment along with the various network interfaces. Although the TWAG and TWAP are shown as separate entities, they can also be co-located in the same physical location for maximum deployment flexibility.

Figure 1-4 TWAG/TWAP Network Deployment



See the *Affirmed Networks TWAP/TWAG Administration Guide* for more information.

ePDG

The MCC *evolved Packet Data Gateway* (ePDG) is a virtualized, software-based Wi-Fi gateway that manages the secure access and transport of data/voice from untrusted non-3GPP access networks, such as open Wi-Fi and public hotspots, to an Operator's network (mobile or fixed network). An *untrusted* network refers to one in which the Operator or service provider cannot authenticate users or control the flow of traffic over the network.

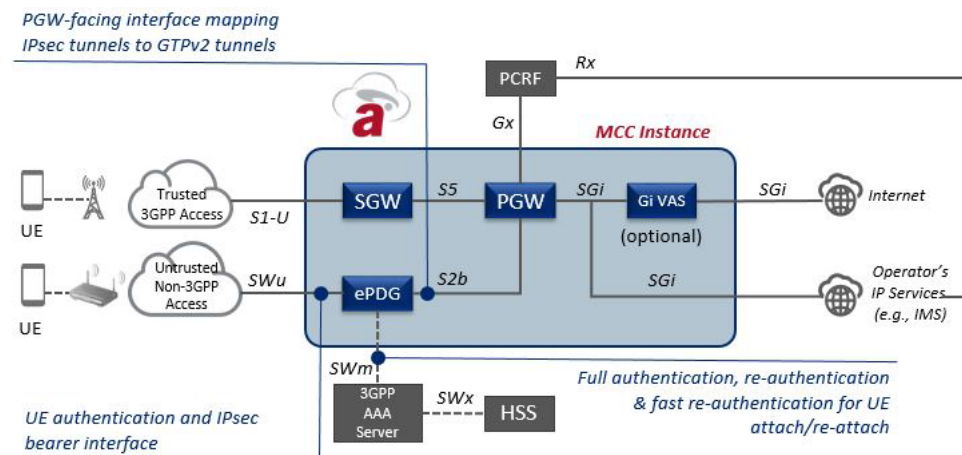
As a robust virtualized software-based gateway, the ePDG delivers 5G levels of IPsec-encrypted communications on a single server. The ePDG uses IPsec tunnels for UE authentication and secure data transport. The ePDG terminates the UE IPsec tunnels and maps them to GTP tunnels to the Packet Data Network Gateway (PGW). Each UE session is securely transported and terminated at the ePDG for secure communication, end-to-end, between the UE and the EPC.

The ePDG also supports seamless handovers from 3GPP access to untrusted non-3GPP access to the EPC. This capability offers mobile Operators the ability to extend their coverage and services to locations where 3GPP access is not yet available.

The ePDG mobile core node is defined in 3GPP specification TS23.402 *Architecture Enhancements for non-3GPP access*.

Figure 1-5 illustrates the position and role of the network components and the MCC ePDG deployment and associated network interfaces.

Figure 1-5 ePDG Network Deployment



See the *Affirmed Networks ePDG Administration Guide* for more information.

Control and User Plane Separation (CUPS)

Affirmed Networks supports Control and User Plane Separation (CUPS) in its virtualized Evolved Packet Core (EPC). CUPS is an LTE enhancement that separates the *Control Plane Function (CPF)* and *User Plane Function (UPF)* into separate nodes. It is defined in 3GPP Technical Specification 23.214 (Release 14) and 3GPP TS 29.244 (Release 14).

CUPS extends LTE by providing options for flexible network deployment through distributed or centralized functions, optimized PDN connections, and independent scaling of functions in the CPF and UPF. To support increased data traffic, Operators can add UPF nodes without adding more CPF nodes.

Figure 1-6 shows the MCC CUPS architecture for the CPF.

Figure 1-6 CUPS Control Plane Function

Affirmed SAEGW-C Cluster Architecture

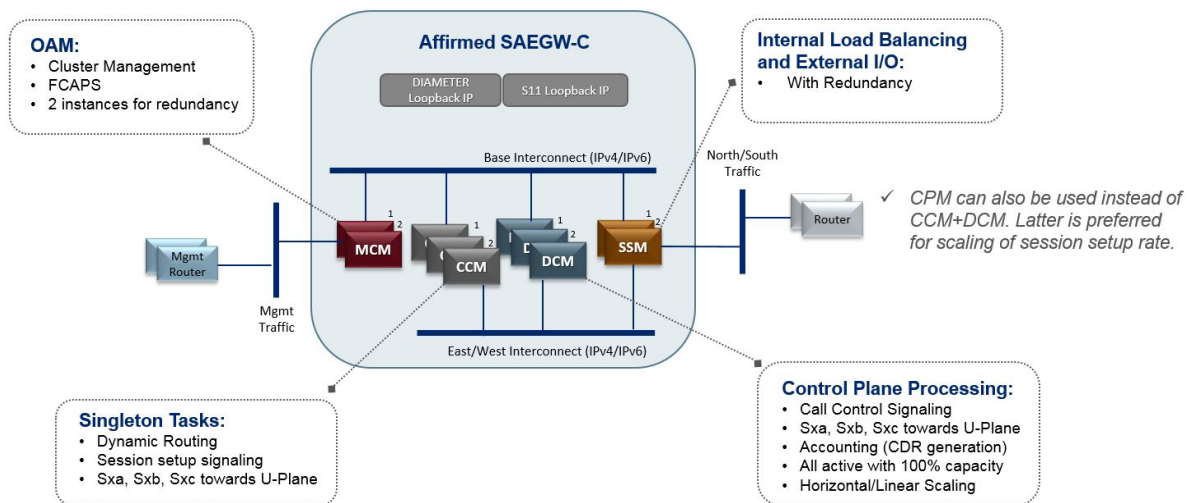
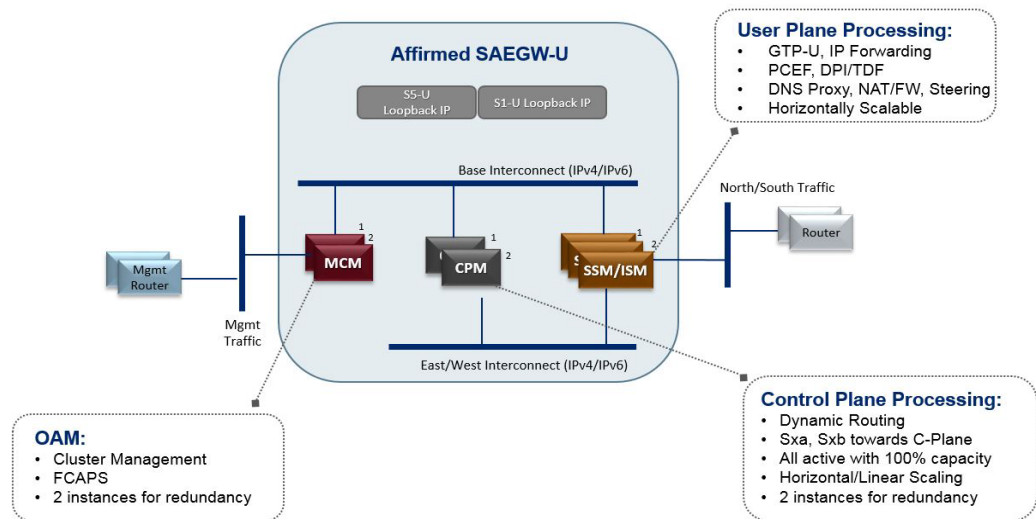


Figure 1-7 shows the MCC CUPS architecture for the UPF.

Figure 1-7 CUPS User Plane Function

Affirmed SAEGW-U Cluster Architecture



Sx Nodes and Interfaces

CUPS separates the Serving Gateway (SGW) and PDN Gateway (PGW) CPF and UPF into the following separate nodes/functions. The C/U nodes are connected through the Sx interfaces.

- Sxa Interface – Standalone SGW-C to SGW-U.
- Sxbc Interface – Standalone PGW-C to PGW-U. The Traffic Detection Function (TDF) is embedded in the PGW-C/U node; it is not supported as a standalone node. The Sxc interface functions as part of the Sxb interface.
- Sxabc Interface – Collocated SAEGW-C to SAEGW-U.

Note: The SAEGW is a combined PGW/SGW.

Packet Forwarding Control Protocol

CUPS uses the Packet Forwarding Control Protocol (PFCP) signaling protocol for Sx interface communications between the Sx nodes. CUPS/PFCP provides:

- Reliable delivery/retry of PFCP messages to peers, including PFCP Session Establishment Request/Response and Modification Request/Response, Report from UPF and Report Response messages

- PFCP interface message-level statistics for CPF and UPF
- PFCP Sx protocol logging which captures the node-related messages and session-related messages sent or received by the PGW-C, SGW-C, PGW-U and SGW-U (CPF and UPF)

For more information about CUPS and PFCP, see the *Affirmed Networks Gateway Administration Guide*.

5G Non-Standalone New Radio (5G-NSA NR) Dual Connectivity

The Affirmed Networks 5G Non-Standalone New Radio (5G-NSA NR) solution supports 5G networks and devices over existing 4G infrastructure. 5G-NSA NR dual-connectivity allows 5G-enabled devices to connect using 5G frequencies over 4G LTE networks for improved data throughput, while using 4G for non-data communications such as through cell towers and servers.

The Affirmed 5G-NSA NR solution leverages the Evolved Packet Core (EPC) network to provide a cost-effective path to 5G. The 4G LTE network provides control functions for the 5G NR. The increased coverage for the NR boosts user plane capacity during peak/high volume periods.

The Affirmed Networks 5G-NSA NR standalone solution is the first of a three-part 3GPP-defined evolution to full 5G implementation. This 5G-NSA NR solution leverages the Evolved Packet Core (EPC) network to provide a cost-effective path to 5G. The 4G LTE network provides control functions for the 5G NR. The increased coverage for the NR boosts user plane capacity during peak/high volume periods.

The second phase upgrades existing Evolved Node Bs (eNBs) to support both NextGen Core and EPC interfaces. In the third phase, the upgrade path provides for connecting a 5G NR to a Next Generation 5G Core.

For more information, see the *Affirmed Networks Gateway Administration Guide*.

Acuitas Service Management System

Acuitas™ Service Management System is the application that Operators use to navigate, monitor, and manage mobile functions and services. Acuitas features a powerful workflow engine and service orchestration capabilities. Additionally, it provides complete control of operational parameters, such as instantiation of Virtual Machines (VMs), status indicators, logs, alarms, KPIs, reports, and dynamic capacity scaling.

The Acuitas Service Management System enables wireless network Operators to manage single or multiple systems using a standard, web-based management interface. See [Chapter 2, Acuitas Service Management System](#) for more information.

Command Line Interface

The Command Line Interface (CLI) enables Operators to enter commands to perform specific tasks to configure, control, query, and monitor system hardware and software components. See the *Affirmed Networks CLI Reference Guide* for more information.

CLI commands are strings of commands or keywords and user-specified arguments that set or modify specific system parameters. Commands are grouped by function and the various command modes with which they are associated. CLI commands are parsed in a top-down manner from left to right and contain standard command line editing, command line history, tab completion, and context-sensitive Help messages.

AI-vProbe

The Affirmed Active Intelligent vProbe (AI-vProbe) is an intelligent tool that provides the following features:

- **Event Data Record (EDR) Generation:** Data reporting mechanism that provides timely activity and statistics information about PDN sessions and subscriber requests. EDRs provide visibility into session, bearer, flow (data plane connection), transaction state, control plane procedures, and SGSN and MME transaction data.
- **Control Plane Traffic Mirroring:** Protocol logging with a GRE-based sink to replicate control plane traffic. vProbe adds a GRE header and sends the encapsulated packet to a remote destination.
- **Data Plane Traffic Mirroring:** Subscriber data plane traffic mirroring. A configure vTap object replicates subscriber traffic and either streams the replicated packets or logs them to a file. The subscriber vTap configures:
 - One or more associated subscriber filter(s) that define the set of subscribers whose traffic is to be replicated. With multiple filters, a packet is considered to match the vTap object if it belongs to one of the sessions that match the configured filters.
 - One or more associated sink(s) that define whether to stream the replicated packets or log them to a file. For multiple sinks, the matching packet is replicated and sent to each sink.
 - One or more associated reference point(s) that specify where to take a snapshot of the received subscriber packet. With multiple reference points, a copy of the packet is taken and sent out for each reference point.

See the *Affirmed Networks Active Intelligent vProbe System Administration Guide* for more information.

Mediation Service Function

The MCC AI-vProbe streams event data records (EDR) to the Mediation Service Function (MSF) application over TCP. The EDR records provide visibility into session, bearer, flow (data plane connection), transaction state, and control plane procedure data. The MSF converts the information to CSV file format. The MSF operates as a standalone node or a multi-node MSF cluster and supports the following functions:

- Receives, processes, and stores EDR streams from the AI-vProbe and converts HTTP EDRs to syslog event records.
- Allows redundant (primary and secondary) TCP connections from MCC to MSF.
- Supports configuration of CSV file generation in MSF. The Operator configures which EDR fields to map to the CSV file and which fields to ignore.
- Supports the posting of formatted EDR data to third-party systems such as syslog servers, HTTP(s) recipients, and other types of OSS systems.

Event Data Record Types

The Mediation Service Function supports the following EDR record types:

- SESSION (EDR_SESSION_RECORD)
- BEARER (EDR_BEARER_RECORD)
- FLOW (EDR_FLOW_RECORD)
- HTTP Transaction (EDR_HTTP_RECORD)
- CPP (EDR_CPP_RECORD)
- MME (EDR_MME_RECORD)
- SGSN (EDR_SGSN_RECORD)

These records report the following information:

- Activity data related to PDN sessions and bearers related to those sessions
- Individual data flows established within a bearer
- Various transactions initiated by a subscriber on those flows
- Control plane procedure (CPP) data from GTP, Diameter, and IKE interfaces
- MME and SGSN information

See the *Affirmed Networks Mediation Service Function Administration Guide*.

Acuitas Service Management System

This chapter describes the following Acuitas features and functions:

- [Overview](#)
- [Acuitas Server and Database Platforms](#)
- [Acuitas Capacity and Performance](#)
- [High Availability Systems](#)
- [Disaster Recovery Backup for the Primary Acuitas HA pair](#)
- [Geographic Redundancy Manager](#)
- [Acuitas Northbound Interfaces and Management Modules](#)
- [Navigating the System](#)
- [Topology](#)
- [Configuration](#)
- [Fault Management](#)
- [VNF Manager](#)
- [Tools](#)
- [Security](#)
- [Performance Management](#)
- [Acuitas Applications](#)
- [Channel Bonding on Physical Interfaces](#)
- [Acuitas Authentication, Authorization, and Accounting](#)

Overview

The Acuitas Service Management System enables Operators to manage the Affirmed network devices using a standards compliant, web-based management interface.

Using Acuitas, Operators can monitor faults, alarms, performance statistics, provision and configure services, and perform software updates on any system. The Affirmed Acuitas Server runs on a Linux (Red Hat) based operating system, while the client is accessible from a Firefox browser or Google Chrome browser.

For more information on Acuitas features and functions, see the following guides:

- *Acuitas User's Guide* – Describes how to use the Acuitas Service Management System to perform tasks, such as adding Network Elements and using Topology, Configuration, Fault, Performance, Virtual Network Function Manager, Security, Tools, CUPS Manager, Workflow, and Configuration Navigator applications.
- *Acuitas System Administration Guide* – Describes how to install, configure, operate, and maintain the Acuitas Service Management System.
- *Acuitas Performance Statistics* – Lists the performance statistics viewable through the Acuitas Service Management System.
- *Software Release Notes for the Acuitas Service Management System* - Contains the software release notes for the Affirmed Networks Acuitas Service Management System.

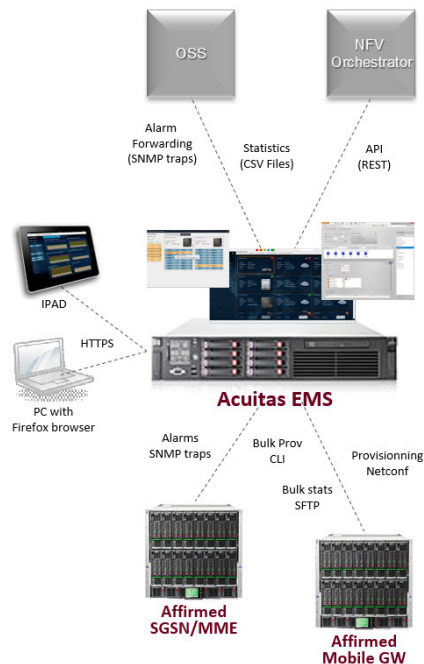
Acuitas Features

Acuitas supports the following features:

- Facilitates efficient deployment, upgrade, and configuration of the system.
- Provides physical and service-level views of the system based on user-access levels.
- Provides reports based on fault, performance, and analytic data collected from the system that can be used to monitor the behavior of the device.
- Supports bulk configuration of the system to help Operators scale their networks.
- Provides an OSS Northbound interface for forwarding SNMP traps to third-party applications such as HP OpenView.
- Performs device-management operations using the NETCONF IETF standards-based protocol.

- Configuration Export Manager provides the ability to save an NE's configuration to raw CLI output. Operators can provision an NE and save its configuration as a "golden" configuration to be input into a configuration template and applied to other NEs.
- Streams the following entities to a remote syslog server:
 - Configuration audit trails
 - Security audit trails
 - Alarms
- VNF Manager controls and monitors virtualized MCC clusters in real-time.
- CUPS Manager application manages Control and User Plane Separation (CUPS) in the MCC

Figure 2-1 Acuitas EMS and VNF Manager



Acuitas Server and Database Platforms

The Acuitas server and database software can be installed on a Bare Metal System or a Virtual Machine.

- HP ProLiant DL380 Gen9
- HP ProLiant DL360 Gen10 (with 64GB, 4 x 300GB drives)
- Blade of the HP BladeSystem c7000

Bare Metal Installation: If the server supplied by Affirmed Networks is for a bare metal installation, RAID 1+0 is used.

VM installation: If the server supplied by Affirmed Networks is for a VM installation, RAID 5 is used.

For Acuitas server hardware specifications, see the *Affirmed Acuitas System Administration Guide*.

Acuitas Virtual Machine Specifications

This section lists the minimum specifications for Acuitas running as a virtual machine.

The following table lists the minimum virtual machine specifications.

Table 2-1 Minimum Virtual Machine Specifications

Specification	Description
Disk Provisioned Size	200GB <i>Note: If you have more storage available, you can increase the provisioned size once the Acuitas package has been deployed.</i>
CPU	16 vCPU <i>Note: If your system has the resources available, you can increase the number of CPUs once the Acuitas package has been deployed.</i>
Memory	32GB

Acuitas Capacity and Performance

This section lists the capacity and performance specifications of the Acuitas Service Management System.

Table 2-2 Acuitas Capacity and Performance

Capacity Metric	Specification
Number of Managed Network Elements	Based on the recommended server configuration, Acuitas can centrally manage up to 100 Network Elements, which can be a combination of MCCs and MCC MMEs.
Traps and Alarm Correlation	Acuitas can process traps and perform alarm correlation at the rate of 100 traps/sec sustained without any loss. This number is likely sufficient for handling traps from 100 or more Network Elements.
Performance Statistics	Acuitas collects bulk Performance Statistics at 5-minute intervals, which provides significant efficiencies over the traditional use of SNMP for such purposes. Collected statistics are stored in the database (up to 2 years) and aggregated into less granular statistics over time. This type of aggregation allows historical trending of performance statistics, while fixing the database size. Each Network Element requires approximately 100 MBs for storage of long term performance data, or approximately 10GB for 100 Network Elements.
Logs	Acuitas collects logs on demand for troubleshooting purposes and purges them automatically after one week, thus capping the disk space requirements.
Configuration Backups	Configuration Backup files are retained for a maximum of 90 days. The file size is typically 30MB total per device, or 3GB for 100 Network Elements.

High Availability Systems

Acuitas offers High Availability (HA) configurations with one Acuitas server and one warm-standby Acuitas server. HA systems are usually *co-located* and allow either *automatic* or *user intervention* switchover from the active to the standby system in case of hardware and/or software failure in the active system. See [Figure 2-2](#) for an example.

One server is configured to serve as the primary server while the other is configured to serve as the secondary server. These roles persist across upgrades.

By contrast, the runtime roles of the servers can vary:

- *Active server* — the server that owns the virtual IP address of the cluster and has the full Acuitas server running.
- *Standby server* — the server in warm-standby mode that can take over in the event that the active server becomes out of service.

Possible triggers that can cause an automatic failover in the HA system include:

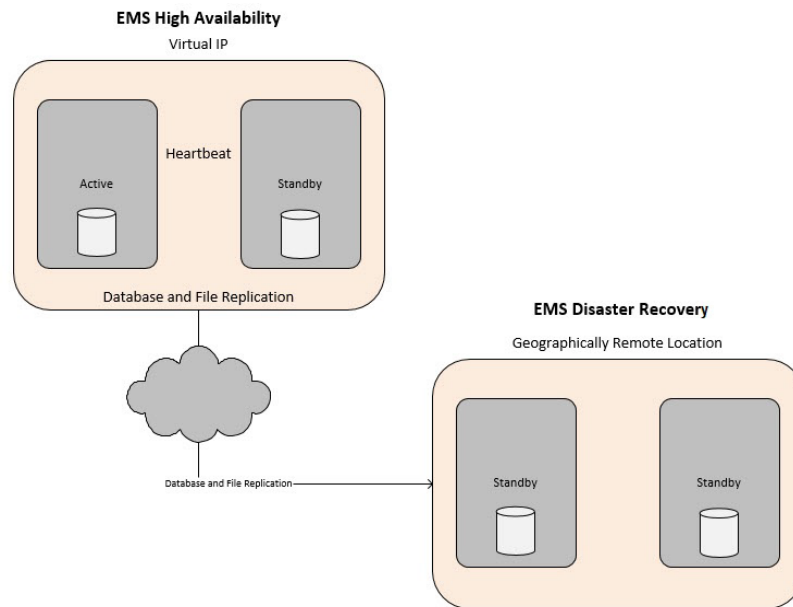
- The active Acuitas server is shut down for some reason, such as because the power supply failed (for example, the rack lost power).
- On the active system, the Acuitas java process hangs, crashes, or otherwise becomes unresponsive.
- The active Acuitas server is taken off the network and cannot be reached by the standby (at which point the standby becomes active).

In addition, Operators can trigger a manual switchover. For example, to perform maintenance operations on the active Acuitas server (such as, upgrading Acuitas software), manually trigger switchover to standby, then bring down the active Acuitas system.

Disaster Recovery Backup for the Primary Acuitas HA pair

Acuitas provides Operators with key management control features to ensure high availability, disaster recovery, and data resiliency.

Figure 2-2 Acuitas Deployments



An Acuitas server can serve as a Disaster Recovery backup for the primary Acuitas HA pair. In a 4-node Acuitas deployment consisting of two HA pairs, one pair acts as the operational HA system while the other pair operates in a standby Disaster Recovery mode.

When the operational HA pair is unavailable due to a disaster or other operational failure at its facility, the geographically-separate standby HA pair is activated to take over Acuitas functions.

Geographic Redundancy Manager

The Acuitas Geographic Redundancy solution is based on a collaborative system of two geographically separated network elements functioning as standby partners for each other. A Geographic Redundancy (GR) group has a designated primary and a designated secondary system. Each group represents a set of services with an active representation on the primary node and a standby representation on the secondary node. If the active node goes out of commission, the services on the standby node can be enabled so these services can resume.

The *Acuitas Geographic Redundancy Manager* is the mediator and facilitator of the Geographic Redundancy solution and enables you to manage and monitor the GR groups. Acuitas mediates the collaboration between nodes and performs user-initiated actions such as replication and switchover.

MCC supports Geographic Redundancy in active/standby and active/active modes:

- **Active/Standby** — Geographic Redundancy operates at the geographic redundancy group level in two MCC clusters located at different geographic locations. One cluster contains the primary/active node, the other cluster contains the secondary/standby node.
- **Active/Active** — Geographic Redundancy operates at the MCC cluster level with the geographic redundancy clusters located at different geographic locations. Each cluster has a primary GR group and a secondary GR group. However, the nodes in the same GR group are located in different geographic-site clusters.

The GR group supports the following high-availability (HA) modes:

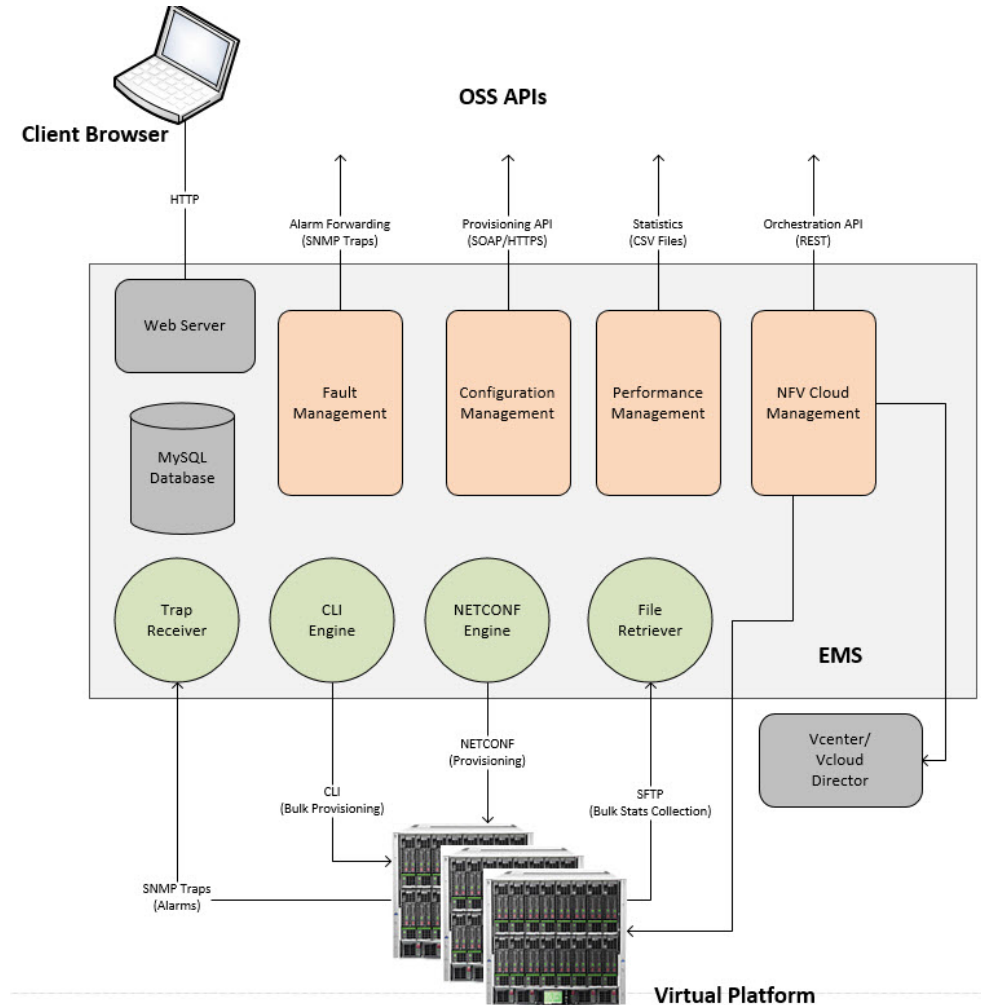
- **warm-manual** — No session synchronization occurs between nodes. Failovers are Operator controlled.
- **hot-auto** — Synchronization of per-session data occurs between nodes. Failovers are automatically triggered based on network health monitoring results. In addition, failovers can be Operator controlled.
- **hot-manual** — Synchronization of per-session data occurs between nodes. Failovers are Operator controlled.

For more information, see the *Acuitas User's Guide*.

Acuitas Northbound Interfaces and Management Modules

Acuitas supports fault management, performance management, provisioning APIs, and orchestration APIs as shown in [Figure 2-3](#). See [Navigating the System](#) for information on each function.

Figure 2-3 Acuitas Northbound Interfaces



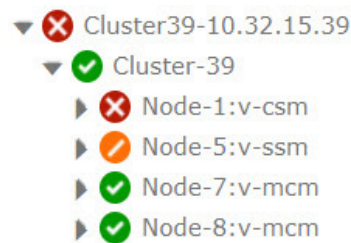
Navigating the System

Acuitas has an easy-to-use graphical user interface with a tree-based representation of the network in the left pane and tabs in the right pane.

Tree Display

The tree display allows you to view devices deployed in different locations. You navigate through objects in the tree much like you navigate through folders and files listed in a file management program. Click the expansion icons to open and close objects in the tree display.

Figure 2-4 Tree Display



Tabs

Tabs on the right panel apply to the Global level and the group level. The data displayed in each tab is filtered based on the elements in the selected group:

Table 2-3 Navigating Acuitas Using Tabs

Tab	Description
Network View	Displays the devices connected to Acuitas in a grid. By default, the Acuitas server is added the first time the server starts up. Each network element is added when you explicitly add it.
Element Details	Lists the details of the devices in the Network View, including Element Name, IP Address, Group, Type, Status, and Reachability.
Alarms	Lists the details of the alarms generated by the network system. Various operations can be performed on the alarms from the tool bar.
Host System Alarms	Visible only when a virtual node or cluster is selected in the tree. This tab lists the details of the alarms generated by blade system components, such as iLO and Host alarms associated with a virtual node.
Event Logs	Lists the details of the events occurring on devices.
PM Dashboard	Displays the Key Performance Indicators (KPIs) collected during the previous 24 hours.

Table 2-3 Navigating Acuitas Using Tabs (Continued)

Tab	Description
System Availability	Displays a graphical representation of a device's Up Time, Mean Time to Failure, Mean Time to Repair, and Mean Time between Failure.

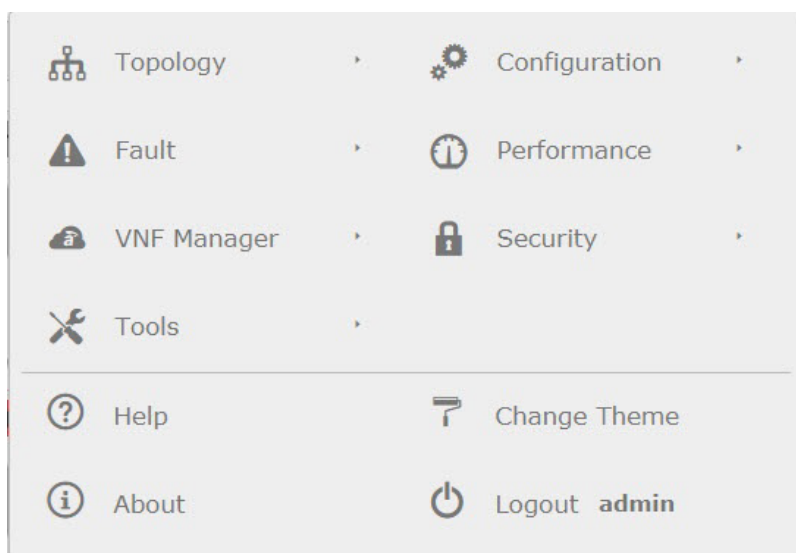
Common Functionality

Acuitas uses common functionality in various areas of the application, such as:

- **Filters** – Customize the details that Acuitas displays based on specific parameters, values, and conditions.
- **Column Properties** – Specify which columns you want displayed and whether you want them to remain displayed so that you do not have to scroll to see them.
- **Record Navigation** – Specify how many records you want displayed per page. Navigate through multiple pages of records.
- Export Records
- Print Records
- Refresh the Screen

This section describes the Acuitas modules that you can access from the toolbar icon as shown in [Figure 2-5](#).

Figure 2-5 Acuitas Modules



Topology

The Topology application provides tools to build a physical and logical topological model of the network. Network elements can be modeled as objects that are stored in the database, thus helping to manage individual or sets of network elements. The Topology applications enables Operators to:

- **Manage Network Elements** - Add, view, modify, delete, and manage network elements.
- **Manage Groups** - Add, modify, and delete groups. A group is a logical collection of network elements. The Global group is a parent group under which all groups reside. You create a group to aggregate network elements for ease in monitoring and control. For example, you can create a group to represent a single customer, or customers located in a city or state, based on the purpose of deployment of the network elements.
- **Configure Audit Trail Purge Settings** - Audit Trail logs the Acuitas activities/events performed by the user. Audit trail is automatically enabled. You can specify how many days you want the Audit Trail saved.
- **Configure Mail Server** - You can set the parameters that specify how the application sends messages to the mail server, including SMTP access and authentication settings. If you configure trap forwarding, the mail server forwards alarms as traps to other specific management systems.
- **Configure Global SNMP Settings** - SNMP parameters are used as global SNMP settings.
- **Configure Message Prompt on Locked User Accounts** - The Administrator can configure the email ID to be included in the system-level message prompt to any user attempting to log into a locked or expired account. For example:
`Please contact J.Doe@XYZCorp with your account request`

Configuration

The Configuration application enables Operators to:

- **Manage Control and User Plane Separation (CUPS)** - Use the CUPS Manager to manage CUPS in MCC devices. For example, you can configure the required Sx interfaces and Packet Forwarding Control Protocol (PFCP) objects on the Control Plane and User Plane, configure geo-redundant Control Planes in a Geographic Redundancy group, and replicate common configuration from the single Control Plane (or active Control Plane in the Geographic Redundancy group) to the User Plane.
- **Manage Software Releases** - Manage the software releases and DPI signature files loaded on the different network elements in the network.

- **Manage Software Upgrade Schedules** - Manage the schedules for upgrading the software releases and DPI signature files of a single and/or multiple elements. For example, you can schedule a software upgrade for a set of Network Elements as part of a typical maintenance cycle. Acuitas can be configured to execute it automatically.
- **Manage Configuration Templates** - Apply similar configurations to multiple elements using configuration templates, while significantly reducing time consuming manual work.
- **Schedule Templates** - Schedule templates to be periodically executed.
- **Manage Audit Trail** - Record all configuration operations and activities.
- **Manage Alarm Triggered Configuration Templates** - Apply configurations to elements based on alarm triggers using alarm triggered configuration templates. Configuration Templates designated as type Alarm Triggered can be specified to be executed when the alarm is received.
- **Manage File Servers** - Use File Servers across a location to maintain network element database backups and network element software.
- **Manage and Monitor Geographic Redundancy Groups** - The Acuitas Geographic Redundancy solution is based on a collaborative system of two geographically-separated chassis functioning as standby partners for each other. A Geographic Redundancy group has a designated primary and a designated secondary system. Each group represents a set of services with an active representation on the primary node and a standby representation on the secondary node. If the active node goes out of commission, the services on the standby node can be enabled so these services can resume.
- **Manage NE Configuration Files** - Use the Configuration Export Manager to save an NE's configuration to raw CLI output. You can provision an NE and save its configuration as a "golden" configuration to be input into a configuration template and applied to other NEs. For instructions on creating configuration templates, see Configuration Templates. You can store up to three configuration files for an NE.
- **URL List Management** - Use URL List Management to manage files of URLs and avoid input errors. A single URL file can contain a maximum of 5 million individual URLs.

Fault Management

Acuitas allows you to view and manage alarms and events, including:

- Identify network elements with alarms
- Configure the fault-management functions that:
 - Detect and isolate malfunctions in a network
 - Maintain logs of traps received from the system for Operators to examine for any network fault
 - Acknowledge and act on error detection notifications
 - Trace and identify faults associated with network elements or the Acuitas system.

VNF Manager

The Virtual Network Function (VNF) Manager allows you to perform full life-cycle management of clusters. You can use the VNF Manager to:

Commission Clusters:

- **MCC Clusters:** Automate deployment of MCC clusters in the following systems:
 - HP BladeSystem c7000 multi-chassis, 16-blade/32-VM deployment (two VMs per blade)
 - HP ProLiant DL380 Gen9 Server

MCC cluster commissioning interoperates with VMware vCenter, vCloud Director, or OpenStack to perform the following function:

- Upload software images
- Instantiate MCC cluster
- Configure MCC cluster for service

- **MCC MME Clusters:** Automate deployment of MCC MME clusters.

MCC MME cluster commissioning interoperates with OpenStack to perform the following function:

- Upload software images
- Instantiate MCC MME cluster
- Configure MCC MME cluster for service

Monitor Clusters:

- Monitor capacity utilization of MCC clusters (*MCC Clusters Only*)
- Report on network elasticity for:

- Life-cycle management events, including cluster instantiation, scale out/scale in, and termination.
- Conditions leading to dynamic scale out/scale in events
- Report on network inventory for:
 - Clusters and their nodes
 - Blade Systems and their blades
 - Server Groups and their servers
 - Hosting relationships between nodes and blades

Scale Out/Scale In Clusters:

- Scale out/scale in network capacity by creating and removing virtual machines (VMs) within a cluster. This process can be performed in the following ways:
 - Manually when initiated by an Operator
 - Dynamically when the capacity is overloaded/underloaded as determined by an associated policy (*MCC Clusters Only*)
 - Programmatically through the VNF Manager and Provisioning API, which is a representational state transfer (REST)-based API

Tools

The Tools application enables Operators to:

- **Administer Database** - Schedule a database backup for automatic execution and trigger a backup manually on demand. The backup can also be used to restore a database from an earlier date and time.
- **Configure Alarm Severity Color Settings** - Configure alarm severity color settings and restore alarm severity color settings to system default settings.
- **Back up Network Configuration** - Perform a back up of configuration data and network inventory reports for each of the reachable managed network elements to individual files. You can schedule the backup of NE configuration data for automatic daily execution and also trigger a backup manually on demand.
- **Back up and Restore Network Element Configuration** - Perform a back up of the NE configuration and store it on the local Acuitas server or a remote server. The backup is available to restore the configuration to a previous known state in the event of lost or incorrect configuration. You can trigger a backup manually on demand and schedule a backup of automatic daily execution. NE configuration backup files are retained for a maximum of 90 days. You can select a successful backed up configuration file and restore it on the NE. The result would be the NE configured exactly as the backup configuration.

Security

Security management allows you to control and monitor user access to the network and associated network management information. Specifically, you control user authentication, user authorization, and user privileges.

Acuitas supports over 200 separate privileges and any set of privileges can be combined to create a security group. Users can belong to multiple security groups.

Administrators use Security applications to:

- Add users and security groups dynamically.
- Assign which managed elements can be accessed by a security group.
- Assign users pre-defined privileges based on user role, such as Administrator, Provisioning, and Maintenance.
- Assign permissions on a granular basis, such as permission to view alarms and assign alarms. Acuitas supports a list of pre-defined permissions.
- Track user activities via Audit Trails.

Security Groups

Security groups logically group users based on user role and define the operations users can perform. For example, members of a customer support organization could be assigned to the Administrator group with administrative privileges. Acuitas uses security group permissions to control group members access to specific types of objects and control the actions the members can perform on those objects:

- Only those management functions granted to the Operator are visible in menu selections.
- Only those objects accessible to a member are visible in the Configuration Navigator.

Acuitas supports the default Administrator and Operator security groups.

- **Administrator** – administrator privileges and can view managed objects and operations. An Administrator can also:
 - Add additional security groups
 - Copy, modify, and delete existing security groups
 - Manage active user sessions
 - Change a user password
 - Unlock users
 - View a user audit trail – Export in CSV, XML, XLS, PDF, HTML formats

- View a list of active users
- **Operator** – view privileges only.

Performance Management

Acuitas contains Performance Management tools that allow you to recognize current or impending performance issues that can cause problems for network users. Activities include the monitoring and maintenance of acceptable network performance, and the collection and analysis of statistics critical to network performance.

Administrators use the Performance application to:

- **Manage Performance Collection and Export** - Use to collect and export PM data, such as:

Raw Performance Management data:

- Acuitas periodically exports Performance Management data to comma-separated values (CSV) files on the Acuitas server.
- You can configure Performance Management reports to be exported to CSV files.
- You can export a historical Performance Management report as textual data or graphical data in various formats, including csv, pdf, xml, xls, html.

KPI data:

- Acuitas periodically exports KPIs to comma-separated values (CSV) files on the Acuitas server.
- You can export KPIs as a daily report.

- **Define Performance Statistic Thresholds** - For SSF, Acuitas Performance Manager also supports configuring Thresholds with various severity levels for a performance statistic attribute or a KPI. When the data collected for an attribute or KPI crosses the threshold level defined, a threshold event is generated with the specified severity.
- **Manage Users Tasks Scheduler** - Use to trigger data collection tasks based on task templates to be periodically executed.
- **Manage Users Tasks Templates** - Use to collect the metrics for performance statistics.
- **Configure Performance Reports** (see [Performance Reports](#)).
- **CSV Stats Recovery** - Perform on-demand statistics recovery in MCC devices.

Performance Reports

Acuitas provides historical, real time, and outage performance reports. Acuitas supports various types of statistics, for example:

- System:
 - CPU Utilization
 - Memory Utilization
- Data plane:
 - Uplink and downlink data and packet rate per interface
 - Uplink and downlink data rate, overall and per session or PDP context
 - Uplink and downlink packet drop rate
 - Average uplink and downlink packet size
- DPI:
 - Overall Detection and decode byte and packet counts
 - Per application byte and packet counts
 - HTTP GET request and response counts, and average service times
 - Top-N applications, protocols
- Control plane:
 - Session / PDP context create and delete success rate
 - PDP context activation / deactivation success rate
 - RADIUS authentication and accounting success rate
 - DIAMETER success and failure rates
 - IP pool utilization per IP sub pool
- Content Services:
 - HTTP proxy uplink and downlink throughput
 - HTTP proxy success rate
 - HTTP proxy transactions per second
 - HTTP proxy optimizations savings
 - TCP splicing rate limiting stats
- TCP:
 - Number of TCP sockets in use
 - Number of orphan TCP sockets
 - Number of TCP sockets established

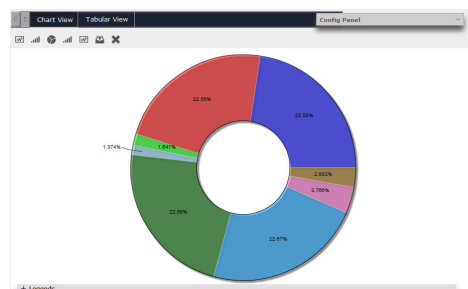
- Number of Time Wait TCP sockets
- TCP memory currently in use (KB)
- TCP memory peak in use (KB)
- TCP event number of sent sync-cookies
- TCP memory low threshold (KB)
- TCP memory pressure threshold (KB)
- TCP memory high threshold (KB)

Historical Statistics

Historical statistics report on the data collected for a specified period. It is a view of the historical data for the performance statistic attributes configured in the data collection tasks for devices.

Acuitas provides the user a view of the historical data of performance statistic attributes for a specified period. Users can display statistics in line, bar, pie, stacked bar chart, and stacked area chart format. You can save the graphical chart as a .pdf file. You can save the textual historical report in various formats, including .csv, .pdf, .xml, .xls, .html.

Figure 2-6 Historical Statistics



Example Chart View

Example Tabular View

chassis11_CPU Utilization_11_2_1	chassis11_CPU Utilization_11_5_1	chassis11_CPU Utilization_11_5_2	chassis11_CPU Utilization
6.08	88.26	88.24	8.21
6.01	88.24	88.23	8.98
6.16	88.24	88.23	8.25
5.99	88.25	88.24	8.14
6.13	88.24	88.24	8.43
5.99	88.25	88.23	8.3
6.11	88.25	88.23	8.5
6.08	88.23	88.24	8.4
6.23	88.26	88.23	8.84
6.21	88.24	88.22	8.94
6.02	88.24	88.23	8.77
6.04	88.25	88.22	8.91
6.02	88.25	88.22	8.47
6.08	88.26	88.24	8.61
6.14	88.27	88.23	8.29
6.11	88.24	88.23	8.56
6.1	88.24	88.23	8.33

Statistics are collected in 5-minute intervals and this data is rolled up as follows:

- 5-minute data is rolled up into 15-minute data after 4 hours.
- 15-minute data is rolled up into hourly data after 24 hours or 1 day.

- Hourly data is rolled up into 3-hour data after 7 days.
- 3-hour data is rolled up into daily data after 90 days.
- Daily data is rolled up into weekly data after 3 years.

For each interval:

- Counter statistics are displayed as per-second average values.
- Gauge and average statistics are displayed as average values.
- Minimum and maximum statistics are displayed as actual values.

Real Time Statistics

Real Time statistics is a live data collected report. It is a graphical view of the current data for the performance statistic attributes configured as a part of the data collection task. Acuitas provides the user a quick graphical view of the current values of performance statistic attributes. The data collected as a part of the Real Time Stats is not persisted. Users with sufficient privilege can start an on-demand polling to monitor performance metrics. Acuitas retrieves the data from the Acuitas database and displays the selected chart in the chart area. The chart is refreshed and redrawn whenever new data is collected based on the server polling interval. Users can display Real Time statistics in line, bar, pie, stacked bar chart, and stacked area chart format.

Outage Report

Outage Report reports on the outage of network elements. An outage refers to the time during which a network element is unreachable. Acuitas tracks the availability of each network element it manages by recording the start and end time for each unreachable window. This report determines the device Lost Time and Regained Time for each NE being managed.

For a list of performance statistics, see *Acuitas Performance Statistics*.

Key Performance Indicators

The PM Dashboard is an Acuitas application that displays the Key Performance Indicators (KPIs). You can save KPI data as textual data or graphical data in a .pdf file. You can also export KPI data to a .csv file.

Acuitas supports various KPIs for performance monitoring, for example:

- EMS PM Stats: displays data flow to and from the Acuitas server
- System KPIs: metrics on system characteristics
- SGW KPIs: metrics on SGW characteristics

- PGW KPIs: metrics on PGW characteristics
- GGSN KPIs: metrics on GGSN APN characteristics
- RADIUS KPIs: metrics on RADIUS characteristics
- APN KPIs: metrics on APN characteristics
- Content Service: metrics on HTTP proxy and caching characteristics
- GTP Transactions: metrics on GTP transactions
- DPI KPIs: historical metrics on DPI characteristics (traffic usage)
- MMS KPIs: metrics on MMS characteristics
- Proxy Buffer KPIs: metrics on packet buffer utilization

For a complete list of KPIs, see the *Acuitas User's Guide*.

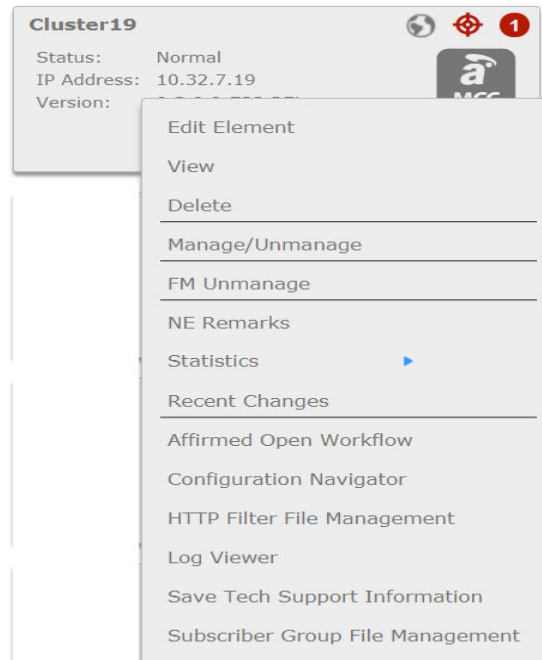
Acuitas Applications

Acuitas contains the following applications on an NE:

- **Affirmed Open Workflow** – Defines the treatment of a service flow across various services offered by the system.
- **Configuration Navigator** – Defines configuration parameters on the system.
- **HTTP Filter File Management** – Defines control lists to either allow or deny content requests to a specific list of network addresses.
- **Log Viewer** – Monitors the system.
- **Save Tech Support Information** – Collects detailed debugging information about a Network Element (such as memory snapshot files and logs).
- **Subscriber Group File Management** – Configures a group of subscribers whose traffic is rate-limited to a configured aggregate QoS Policy. When the aggregate usage of all the users in the group exceeds the configured limits, the packets from these users are dropped.

You access these applications by right-clicking an NE and selecting the application from the menu as shown in [Figure 2-7](#).

Figure 2-7 NE Applications



Channel Bonding on Physical Interfaces

Channel bonding enables multiple network interfaces to act as a single interface, thereby increasing bandwidth and providing redundancy. You can configure channel bonding to operate with Acuitas servers (standalone and High Availability) to remove a single point of failure in the network. See the *Affirmed Acuitas System Administration Guide* for more information.

Acuitas Authentication, Authorization, and Accounting

This section describes the following Acuitas AAA/AA methods:

- [TACACS+ Authentication, Authorization, and Accounting](#)
- [LDAP Authentication and Authorization](#)
- [RADIUS Authentication and Authorization](#)

TACACS+ Authentication, Authorization, and Accounting

Terminal Access Controller Access Control System Plus (TACACS+) is a protocol based on a client-server model where the Acuitas server (a NAS entity) sends requests to and receives responses from the TACACS+ server.

When a user attempts to log into Acuitas, Acuitas sends an authentication request to the TACACS+ server and the server either accepts or rejects the request based on information in its user database.

Upon successful authentication, the Acuitas server sends an authorization request to which the TACACS+ server responds with the user's group assignment (used to determine the commands to which the user has access). The authorization information propagated from the TACACS+ server is via a custom AVP in the form of **group=<user_group>**, where `user_group` is defined in Acuitas. After the session is established, the system sends user accounting messages to the TACACS+ server, including session termination.

To use TACACS+ authentication and authorization, you configure and enable TACACS+ authentication before starting the Acuitas Application.

LDAP Authentication and Authorization

The Acuitas Security Module can enable the LDAP login module and perform the following authentication and authorization:

- When a user attempts to log in, Acuitas connects with the LDAP server for authentication.
- Following successful authentication, Acuitas retrieves user-group mapping details from the LDAP Authentication response.
- Authentication creates a logical user details entry in the Acuitas database to handle user-specific Acuitas operations.

The following sections describe how the LDAP login module supports authentication and authorization in Acuitas.

Authentication

Acuitas authenticates users by checking the list of enabled login modules in the order in which they were configured. LDAP login is enabled in the following order:

- 1 LDAP login
- 2 DB login

When a user attempts authentication, the LDAP login module acts as standard LDAP client. The following high level actions are performed:

- 1 LDAP login module sends the user name and password to LDAP server and attempts to gain access to the user's LDAP context in the LDAP server.
- 2 LDAP server authenticates the user and provides access to the LDAP context.
- 3 Acuitas either creates the Acuitas user session or fails the user authentication based on the response from LDAP server.

Acuitas supports configuring primary and secondary LDAP servers. In the case where the primary LDAP server is not reachable, the secondary LDAP server can be used to authenticate the user.

Authorization

The Acuitas Security Module supports the following user authorization:

- In Acuitas, user operations have individual role permissions.
- Any created Acuitas user is associated with one or more a user groups.
- A set of role permissions is provided for each user group.

Separate user-group mapping and group-role mapping are stored in the Acuitas database.

Following successful authentication, Acuitas retrieves the user group and role mapping details from the Acuitas database. The Acuitas user is permitted to perform only the operations for which they have role permissions in their belonged user groups.

When LDAP based authorization is enabled:

- Following successful LDAP authentication, user-group mapping is expected from LDAP context access provided by the LDAP server.
For example, once a LDAP user named *test* is authenticated in the LDAP server, its user-group mapping *test – Operators* must be provided as LDAP response.
- Acuitas creates a logical user entry in the Acuitas database to handle authorization internally upon new LDAP user authentication attempts. Acuitas updates the group associations with every LDAP login.
- Using the user-group mapping from LDAP and the group-role mapping in the Acuitas database, user authorization functions the same as the existing DB-based authorization.

RADIUS Authentication and Authorization

RADIUS (Remote Authentication Dial-In User Service) is a network protocol that provides centralized authentication and authorization support for an organization.

Acuitas supports RADIUS PAP (Password Authentication Protocol) based authentication and authorization. If the user name is found in the RADIUS database and the password is correct, the RADIUS server returns an Access-Accept response including pre-defined parameters (attribute-value pairs) that grant proper access to the user. These parameters are configured in RADIUS, including service type, protocol type, IP address to assign the user, an access control list (ACL), and other values.

By default, RADIUS based authentication is disabled. To use RADIUS authentication and authorization, you configure the Authentication provider and other settings and enable it following Acuitas installation and before starting the Acuitas Application.

Note: In addition, you add the Acuitas server as an authorized client of the RADIUS server. Do so for all Acuitas servers in the HA or HA+DR cluster.

Acuitas Security Module Authentication Order

The Acuitas Security Module performs Authentication and Authorization of users by implementing the enabled login modules in the following order:

- 1 TACACS+ login
- 2 LDAP login
- 3 RADIUS login
- 4 SANE login
- 5 DB login

Any user attempting login into Acuitas is authenticated using these enabled login modules. In case of any authentication failure in a module, the process proceeds to the next enabled login module. Modules are called in the order defined in the AuthenticationServer.xml file.

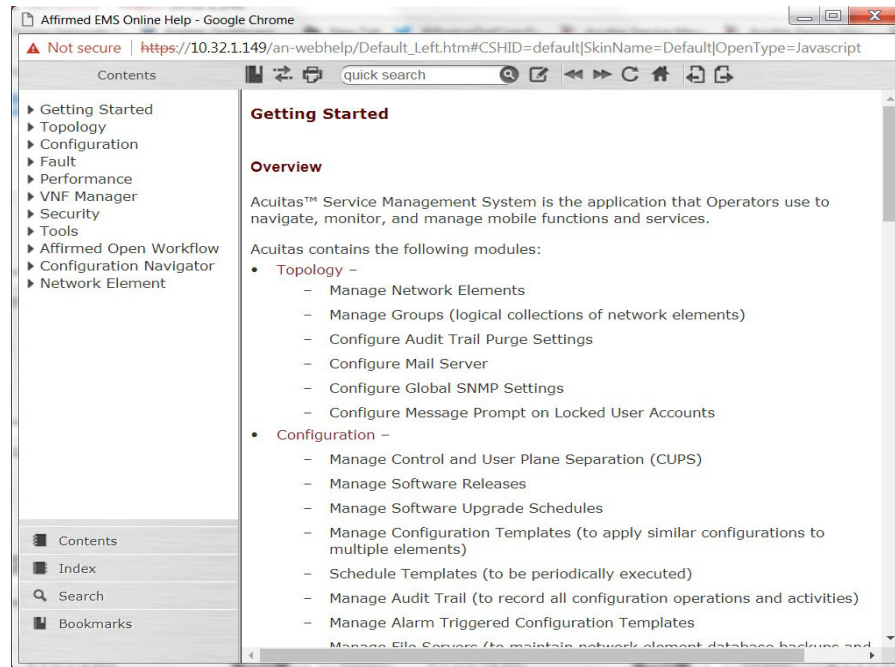
Note: If DB Login is disabled, you must create the `admin` user in the `Administrator` user group in TACACS+/LDAP/RADIUS.

For more information about Acuitas AAA and configuration, see the *Acuitas User's Guide*.

Online Help

Acuitas provides online Help. You can access online Help from the **Help** icon on the toolbar or click **Help** or the **Help** icon from any dialog box. Acuitas displays Help in a separate window, showing the context-sensitive help for Acuitas. See the *Acuitas User's Guide* for information.

Figure 2-8 Top-level Help Window



MCC Virtualized Hardware Environment

This section provides an overview of the Affirmed MCC virtualized hardware environment. See [MCC Software Architecture](#) for information on deployment strategies. For information about your specific configuration, network planning, and hardware requirements, see the guides listed in the *Affirmed Networks Documentation Library*.

The MCC provides services in a virtualized environment, with software executed on multiple virtual machines that are separate from the underlying hardware resources. The virtualized environment includes the following components:

- **Virtual Machine (VM)** – An isolated virtualized hardware platform that provides a configurable amount of CPU processing power, memory, disk space, and network interfaces.
- **Hypervisor Layer** – The VMware ESXi, KVM, or OpenStack hypervisor that provides tools to dynamically provision virtual machines and configure CPU, memory, disk, and network resources assigned to the virtual machine.

***Note:** The OVAs released by Affirmed Networks deploy SCSI drives by default.*

- **Hardware Layer** – An x86 architecture data center server platform composed of computing hardware, virtualization support, switching fabric, and management software hypervisor layer. This layer can be a blade server or a collection of commodity PC servers.

MCC Virtual Platforms

The MCC virtual platforms offer state-of-the art high performance, high capacity, and cost-effective use of hardware and virtual resources. MCC supports VMware ESXi, KVM, and OpenStack platforms. The following sections describe some of these platforms and features.

Virtual Connect vSwitch

The vSwitch configuration is recommended for low/medium bandwidth intensive network and applications (such as M2M). This configuration interfaces the MCC application on top of a software-based switch that isolates the application from the hardware environment. This configuration supports a number of management tool functions, such as VMware vMotion, to move applications without the need to reconfigure IP addresses.

High-performance Direct Single Root I/O Virtualization (SR-IOV)

The SR-IOV configuration is a PCI industry standard and is recommended for high capacity networks and applications. This configuration uses network adapters (NICs), such as Niantic, Intel, Mellanox, and Fortville partitioned into multiple NICs. Each partition has its own PCI registers, queues, and so on. Virtualization occurs within the PCI bus directly, enabling a direct MCC application interface with the NIC driver interfaces. The hypervisor provides the mechanisms to contain the MCC, but the MCC has direct control of the NIC. Using SR-IOV, multiple Virtual Machines (VMs) can share PCI hardware resources. Without SR-IOV, only one VM can run direct IO.

SRIOV Considerations

For highest performance, all North/South and East/West vNICs on all VM types can be configured for SR-IOV (single-root I/O virtualization). SR-IOV is similar to PCI pass-through with the additional capability of multiplexing a single physical NIC (pNIC) across multiple VMs. This configuration allows multiple VMs on the same server to derive all of the benefits of direct physical device management simultaneously. Architecturally, SR-IOV splits the device driver into the following sections:

- *Physical Function (PF)* – This smaller section of SR-IOV resides within the host hypervisor and handles mostly initialization and physical layer aspects of the device.
- *Virtual Function (VF)* – This larger section of SR-IOV is realized as either a guest O/S installed network device driver or a user-space resident Poll-Mode-Driver (PMD).

The user plane VM types (vASM, vIOM, vWSM and the combined types vISM, vSSM, and vCSM), when deployed with SR-IOV on the East-West and/or North-South vNICs, utilize DPDK and therefore execute the Virtual Function Poll Mode Driver for the relevant physical device within the appropriate MCC user space processes. In this deployment scenario, the involved physical network device types must support DPDK Poll Mode Virtual Function Drivers (via the open source DPDK on dpdk.org). Affirmed Networks has integrated those specific drivers into the MCC application. Examples of supported physical device types include the Intel Niantic and Fortville families of MAC devices and the Mellanox family of MAC devices.

The control plane VM types (vCCM, vDCM, and the combined type vCPM), when deployed with SR-IOV on their East-West vNICs, utilize the traditional guest kernel resident, interrupt-based VF driver for the relevant physical device. In this deployment scenario, the hypervisor must have explicit support for the physical device, including support of the VF/PF split for that device type. Typically, the list of supported device types within the hypervisor is a broad superset of those supported by DPDK.

When SR-IOV is used for North/South and/or East/West vNICs, the Base interconnect vNICs are unaffected and continue to use a vSwitch/vRouter appropriate for the selected hypervisor.

SR-IOV/non-SR-IOV Management/Base Network Port Assignments

MCC base devices can be either SR-IOV or non-SR-IOV, even if the management device is non-SR-IOV. The Eth0 and Eth1 Ethernet port mapping of network devices (automatically configured) depends on whether the management device is SR-IOV or non-SR-IOV. When all network devices (including the management device) are SR-IOV, Eth0 is assigned to the Base Ethernet port, and Eth1 is assigned to the Management port. However, when the management device is non-SR-IOV and the remainder of the devices are SR-IOV, the Base/Management port assignments are reversed. In that case, Eth0 is assigned to the Management Ethernet port and Eth1 is assigned to the Base port.

Unbonded Base Ports

In redundant MCC systems, Base and Base 2 Ethernet ports are not bonded. Each separate base port is assigned a unique MAC address.

DPDK-Enabled vSwitch/vRouter Considerations

Within the KVM/OpenStack ecosystem, an emerging group of DPDK-enabled vSwitches/vRouters close the performance gap between traditional vSwitch performance and SR-IOV/PCI pass-through performance, while providing the SDN benefits expected of an overlay approach to data center LAN design. These

vSwitches/vRouters (for example, OVS, Open Contrail, and Wind AVS) use the DPDK Poll Mode Driver and PCI pass-through to bypass the hypervisor and achieve high performance. The supported physical device types are supported by DPDK open source and integrated by the particular vSwitch/vRouter implementation.

A DPDK-based application that runs on top of the DPDK-enabled vSwitch/vRouter uses the DPDK Poll Mode virtio driver as the universal way to integrate to all vSwitch and vRouter types within the KVM/OpenStack ecosystem (that is, DPDK-enabled and traditional interrupt-based vSwitches/vRouters). However, virtio, even with polling, is relatively slow due to a required full-packet copy. Many of the DPDK-enabled vSwitch implementations also make available a specialized zero-copy DPDK Poll Mode driver that the application (for example, MCC) integrates for higher performance.

Note: For details about the supported MCC platforms including third-party software, see the latest Software Release Notes for the Mobile Content Cloud.

MCC Software Architecture

This section provides an overview of the software services that run on different Virtual Machines (VMs).

Overview

The Affirmed Networks MCC is characterized as a cluster of differentiated VMs that collectively provide a single Composite-VNF (C-VNF). The Composite-VNF functions as a single managed Network Element (NE) and has the ability to simultaneously function as any one or more of the individual VNFs provided by Affirmed Networks. These VNFs include access termination capabilities such as PGW, SGW, GGSN, T-WAG, and ePDG as well as value-added network functions such as NAT, Firewall, multi-protocol/HTTP proxy, HTTP content filtering, video optimization, and TCP optimization.

Individual VNFs are activated through configuration and licensing. Individually enabled VNFs do not map to particular VM instances. Instead, all VM instances cooperatively support all of the enabled VNFs simultaneously.

VM Types and Personalities

MCC supports the following VM types that can be specialized into personalities for larger deployments:

VM Type	Description
Management Configuration Module (vMCM)	■ A Management Configuration Module VM instance running core processes, including node and configuration functions ■ Two vMCMs are required per cluster ■ OA&M, CLI, NETCONF, and so on ■ CCM – Centralized Control Module ■ Dynamic Routing, Session management, Diameter
Content Services Module (vCSM) ■ vCCM ■ vDCM	■ A Content Services Module VM instance running core content service functionality, including control/subscriber management processes and infrastructure tasks (statistics collection, alarms, events, and so on) ■ vCSM is a combination of the vCCM, vDCM, and vASM (not recommended for new deployments) ■ GTP-C, call control, RADIUS clients, IKE v1/v2
Subscriber Services Module (vSSM) ■ vIOM ■ vWSM	■ A Subscriber Services Module VM instance running Datapath processes and supporting infrastructure services ■ One vIOM (Input Output Module) – Specialized personality of the Subscriber Services Module (vSSM) – Logical IP interface termination, IPSEC ■ One vWSM (Workflow Services Module) – Specialized personality of the Subscriber Services Module (vSSM) ■ Subscriber/APN tunnel termination ■ QoS, Charging, NAT/Firewall, Policy Routing, DPI ■ IPSEC (per-subscriber – ePDG)

Node Personalities

Node personality defines which services are started on a given node in a cluster. A node can be configured to use a specific personality, each of which runs a predefined collection of functionality sets. This feature allows Operators to run specialized card/nodes with the exact services they want and increases MCC performance.

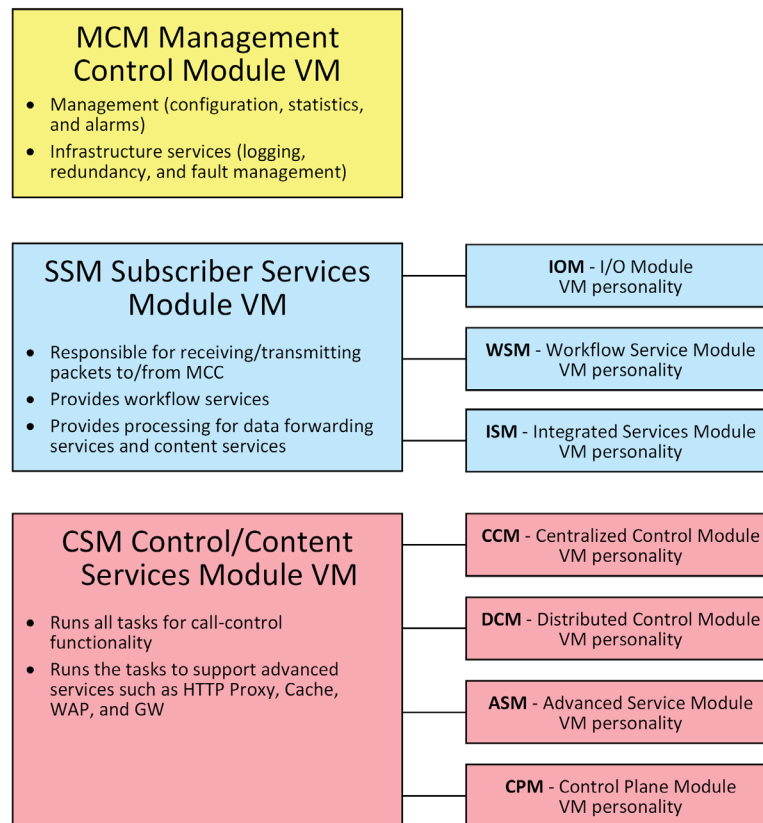
The following table describes the personalities.

Table 4-1 Personalities

Personality	Description
CCM: Centralized Control Module <i>Note: Do not set Admin State to disabled for CCM personality.</i>	<i>Applies to node type vCSM</i> Runs the centralized tasks needed for call-control functionality and routing protocols.
DCM: Distributed Control Module	<i>Applies to node type vCSM</i> Runs the distributed tasks needed for call-control functionality (mainly the PGW, SGW, and RADIUS tasks).
CPM: Control Plane Module <i>Note: Do not set Admin State to disabled for CPM personality.</i>	<i>Applies to node type vCSM</i> Runs all tasks needed for call-control functionality.
IOM: I/O Module <i>Note: Do not set Admin State to disabled for IOM personality.</i>	<i>Applies to node type vSSM</i> Is responsible for receiving packets into the MCC and sending the packets out.
WSM: Workflow Services Module	<i>Applies to node type vSSM</i> Provides workflow services.
SSM: Subscriber Services Module	<i>Applies to node type vSSM</i> Runs all services provided by IOM and WSM.
ASM: Advanced Services Module	<i>Applies to node type vCSM</i> Runs the tasks needed for providing advanced services like HTTP Proxy, Cache, WAP GW, etc.
CSM: Control/Content Services Module	<i>Applies to node type vCSM</i> Runs all services provided by CPM and ASM.
ISM: Integrated Services Module	<i>Applies to node type vSSM</i> Runs content and subscriber services

Figure 4-1 shows the various VM/node personalities and functions.

Figure 4-1 VM Personalities and Functions



Building Block Strategies

When configuring personalities, you first specify a *Building Block Strategy* for the cluster. The strategy specifies the supported combinations of node personalities to build the cluster. The Building Block Strategies and their contents are as follows:

- **Compact Strategy** — For traditional MCM, SSM, and CSM nodes.
 - MCM (Management Control Module)
 - SSM (Subscriber Services Module)
 - CSM (Control/Content Services Module)
- **Advanced-services-separation Strategy** — Optimized for small clusters (less than 12 nodes). This strategy separates the advanced services from the Call-Control Management services (PGW, SGW, and GGSN). Systems that do not provide advanced services *should not* deploy the ASM personality node in this strategy. Systems that do provide advanced services do need to deploy the ASM personality node.
 - MCM (Management Control Module)

- SSM (Subscriber Services Module)
- ASM (Advanced Services Module – Content services)
- CPM (Control Plane Module)
- **Control-services-separation Strategy** — This strategy separates the centralized and distributed aspects of the Call-Control Management services. If either the advanced services or the Call-Control Management services (PGW, SGW, and GGSN) increase over time, the cluster can be expanded/scaled to support the growth.
 - MCM (Management Control Module)
 - ASM (Advanced Services Module – Content services)
 - SSM (Subscriber Services Module)
 - CCM (Centralized Control Module)
 - DCM (Distributed Control Module)
- **Specialized Strategy** — Optimized for large clusters (more than 12 nodes). In addition to the separation provided by advanced-service-separation, this strategy separates the I/O and the workflow services on the SSM node. In addition, it separates the centralized and distributed aspects of the Call-Control Management services. Systems that do not provide advanced services do not need to deploy the ASM personality node in this strategy. Systems that do provide advanced services do need to deploy the ASM personality node. Applies to cluster nodes only.
 - MCM (Management Control Module)
 - IOM (I/O Module)
 - WSM (Workflow Services Module)
 - ASM (Advanced Services Module – Content services)
 - CCM (Centralized Control Module)
 - DCM (Distributed Control Module)
- **integrated-services Strategy** — Centralizes the execution of data forwarding services (workflow and I/O) and advanced services in one VM and Call-Control Management services (PGW, SGW, and GGSN) in another VM. As a result, the cluster can be expanded/scaled to independently support growth of gateway call control services, workflow, I/O, and content services.

Contents:

- MCM (Management Control Module)
- CPM (Control Plane Module)
- ISM (Integrated Services Module: Content and Subscriber Services)

The ISM provides a one-stop processing node for data forwarding services and content services.

workflow-services-separation Strategy (supports ePDG only) — Combines the required building blocks into one building-block strategy to build the ePDG VMs. This strategy streamlines the ePDG VM creation process.

Contents:

- MCM (Management Control Module)
- CPM (Control Plane Module)
- IOM (I/O Module)
- WSM (Workflow Services Module)

When you configure a node for the cluster, you select the node type and node personality.

■ Node type:

- CSM: Control/Content Services Module
- MCM: Management Configuration Module
- SSM: Subscriber Services Module
- vCSM: Virtual Control/Content Services Module
- vMCM: Virtual Management Configuration Module
- vSSM: Virtual Subscriber Services Module

Node Personality: The personalities available for a node are filtered based on the Building Block Strategy specified for the cluster.

Value-Added Services Guidelines

When deploying Value-Added Services (VAS), use the following VM provisioning guidelines:

- Use one of the following cluster building block strategies:
 - specialized (preferred)
 - advanced-services-separation (small deployments only)
 - compact (small deployments only)
 - integrated-services

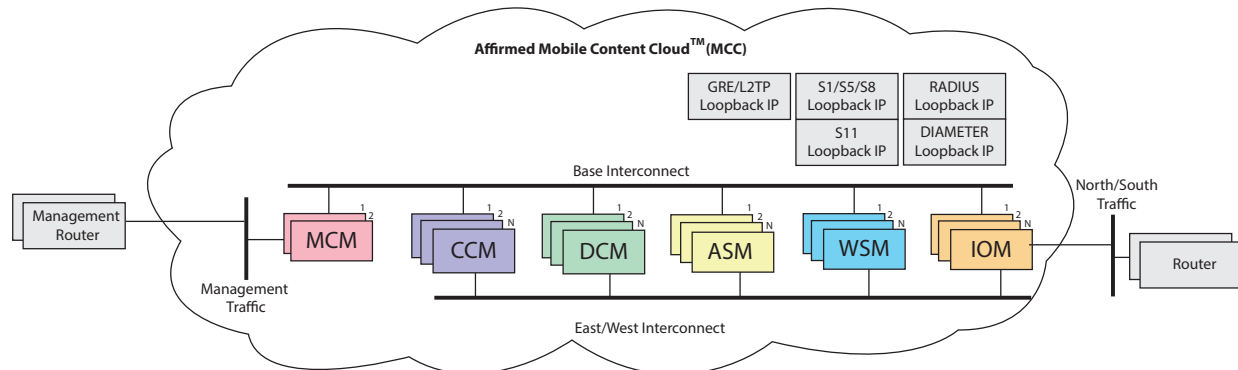
- Provision a sufficient number of CSM VMs as required for your network deployment:
 - use CCM, DCM, and ASM personalities for specialized cluster building block strategy
 - use CPM and ASM personalities for advanced-content-separation cluster building block strategy
 - use default v-CSM personality for compact cluster building block strategy
 - use ISM personality for integrated-services building block strategy
- Designate a minimum of 16 vCPUs for ISMs. Performance and throughput at any vCPU provisioning level (including 16 vCPUs) depends on traffic profiles and call models. Between this assignment and the setting in ASM optimization level, you can scale performance and throughput based on the network services provided in your network deployment.
- Designate a dedicated CPU to each ASM by setting the processor affinity.
- Set the number of ASM virtual cores to the number of CPU cores minus 10%, reserving the remainder for the VM hypervisor. For example:
 - 18 on CPU with 20 virtual cores
 - 22 on CPU with 24 virtual cores

When deploying CPU intensive functionality, the minimum number of virtual cores is 18.
- Provision memory for ASMs or CSM VMs running Value-Added Services:
 - minimum of 55GB for typical deployments
 - minimum of 32GB evaluation deployments with a limited number (up to a few hundred) of subscribers

VM Networking Architecture

The VMs that comprise an MCC require various logical interconnects. These interconnects may be traditional LANs, VLANs, or SDN-enabled emulated LANs. [Figure 4-2](#) provides a high-level view of the cluster interconnect topology for a large deployment model. See [Additional VM Types/Deployment Models](#) for other deployment models.

Figure 4-2 Interconnect Model



- *Base Interconnect* – Used for Inter-Process Communications (IPC) between all VMs. No user-plane or control-plane traffic traverses the base interconnect.
- *East/West Interconnect* – All of the VM types except for the vMCMs transfer user-plane packets, control-plane packets, or both amongst themselves over the East/West Interconnect.
- *North/South Traffic* – Only the vIOMs (Input/Output Modules) send and receive user-plane and control-plane packets outside of the cluster. The North/South vNICs on all vIOMs may reside on a common emulated LAN or may utilize separate point-to-point emulated LANs to interconnect to peer PE routers.
- *Management Traffic* – Only the vMCMs send and receive management-plane packets outside of the cluster. The Management Network vNICs on both vMCMs must reside on a common emulated LAN.

MCC Configuration Models

This section describes the various VM types supported in the following models.

- [Management Plane VM](#)
- [User Plane VM](#)
- [Control Plane VM](#)

Each of the VM types can have multiple instances within the cluster for both scale and resiliency. A cluster may be resized (scale-in, scale-out) either manually or dynamically at any time. All load balancing is handled internally. Peer entities such as MME, PCRF, and RADIUS server are always unaware of the current or future cluster topology.

Management Plane VM

The management plane model is comprised of the vMCM.

vMCM

The vMCM (*virtual Management Control Module*) terminates all external management traffic to its SSH/CLI, NETCONF and SNMP (trap) interfaces. In addition, it hosts certain cluster-level processes that together manage cluster membership, process assignment and distribution, VM liveness determination, and aspects of load balancing and cluster resiliency.

Operators provision an MCC cluster for either one (non-redundant) or two (redundant) vMCM VMs.

Internal Networking Topology

The vMCM requires connectivity from its external-facing vNICs to the OA&M network. If redundant vMCMs are deployed, both must access a shared emulated LAN. The pair of vMCMs utilize a Virtual IP address and gratuitous ARP to achieve failover of the required external connectivity.

In addition, the vMCM must be configured with a minimum of one and a maximum of two Base Interconnect vNICs. Heartbeats are performed over the Base Interconnect(s) with automatic failover, when appropriate.

Dynamic Scaling Considerations

The vMCM is not eligible for dynamic scaling because a maximum of two vMCMs are used within an MCC cluster.

Redundancy

The vMCM handles all FCAPS-related management for the entire MCC cluster. For full resiliency, an MCC cluster should be provisioned with two vMCMs; if one vMCM fails, the other vMCM takes over and provides continuous manageability of the MCC cluster. All persistent file operations, such as CDR storage, are constantly synchronized between the two vMCMs to minimize any potential data loss during a switchover.

The vMCMs each support a single logical IP interface for management purposes. These logical interfaces must be configured on the same subnet. A third IP address configured on the same subnet is used as a virtual IP (VIP) to maintain constant communication with the EMS, OSS, or other northbound applications. Gratuitous ARP is used to fail over the VIP.

Primary/Secondary Relationship

The two MCMs operate in a primary/secondary relationship (rather than an active/standby relationship).

MCC performs a switchover only on the active task that crashed (or a group of dependent tasks). The remaining tasks and the node itself are not affected. The primary MCM is the MCM on which the active GRD task is running. The secondary MCM has the standby GRD task running, but it can have other director tasks running in an active state.

If a standby MCM task crashes, only the crashing standby task is restarted. The remaining tasks and the node are not affected.

Each task running on the MCM is assigned to one of the following recovery actions:

taskFailover – If a task with this recovery action crashes, then only that task is affected. Certain tasks belonging to this group run in non-redundant mode. That is, they do not have corresponding standby peers. The tasks run as active-active on both MCMs. When such a task crashes, MCC restarts this task in active mode. MCC takes the following actions during a task crash:

Active task crash: MCC performs a switchover to the standby instance.

Standby task crash: MCC restarts the task in standby mode.

grdGroupFailover – These groups of tasks depend on each other, therefore, they must be running active on the same node. MCC takes the following actions during a task crash:

Active task crash: MCC performs a switchover on all active tasks collectively, however, tasks outside of this group are not affected.

Standby task crash: MCC restarts only the standby task in standby mode.

The following table maps a task to its recovery action:

Recovery Action	Tasks
grdGroupFailover	FMDirector, HwAgent, ImDir, PolicyListAgent, GRD/NSAgent, PkiDir, Confd, NmdDirector, NmdRestoreServiceDir, NetDirectorCaleaDirector,, LicenseMgr
taskFailover	ContentDirector, CrashAgent, DataRecordDirector, DataRecordAgentMCM, EdrDirector, EventAgent, EventDirector, FileDirector, GeoRedDirector, GwStatsDirector, OFCDiskAgent, PerfStatMgmtDirector, RouteDir, SecDirector, SubscriberTraceEventAgent, TOSGwSd, TOSWfSd
taskFailover (non HA Enabled tasks)	EventConfigRelayAgent, FMAgent, FileAgent, GwStatsAgent, ImAgent, NRA

Since MCMs operate in a primary/secondary relationship, the **show cluster summary** CLI command displays the MCMs as primary or secondary (rather than active or standby).

```
an3000-mcm-slot8cpu1# show cluster summary
```

```
cluster
  SLOT  CPU
  ID NUMBER NUMBER TYPE PERSONALITY MODEL STATE HA STATE STATE CPU UPTIME
  RELEASE
-----
14 2 1 v-csm CSM vCSM enabled active up D:000 H:21 M:33 S:11
6.6.0.0-1088.REL
3 1 v-csm CSM vCSM enabled active up D:000 H:21 M:33 S:11
6.6.0.0-1088.REL
5 1 v-ssm SSM vSSM enabled active up D:000 H:21 M:33 S:11
6.6.0.0-1088.REL
6 1 v-ssm SSM vSSM enabled active up D:000 H:21 M:33 S:11
6.6.0.0-1088.REL
7 1 v-mcm MCM vMCM enabled secondary up D:000 H:21 M:38 S:16
6.6.0.0-1088.REL
8 1 v-mcm MCM vMCM enabled primary up D:000 H:21 M:39 S:55
6.6.0.0-1088.REL

an3000-mcm-slot8cpu1#
```

User Plane VM

The MCC User Plane VM is comprised of the vIOM and vWSM VMs as well as the vASM VM for content services. For compact deployments, the vIOM and vWSM are collapsed in the vSSM VM. The vASM is collapsed in the vCSM VM which also includes the vCCM and vDCM functions.

Note: When a packet fragment arrives on a vIOM or on the IOM function of a vSSM or vISM, it is queued. All the fragments for the packet must be received on the same IOM to successfully reassemble the packet and forward it for additional processing. If the fragments are received on multiple IOMs, reassembly fails and the fragments are dropped. The MCC peer router must be configured to route all of the packet fragments identically.

vIOM

The vIOM (virtual Input/Output Module) provides all input/output towards the external network such as, radio and Internet, and serves as the load balancer of both control-plane and user-plane traffic for the entire cluster. The vIOM is DPDK-based in all deployment scenarios and therefore appears to be always running on most of its vCPUs. Application level KPIs provide the actual utilization of the vIOM (that is, idle time vs. busy time). In addition, the vIOM performs all site-to-site IPSec tunnel processing.

A cluster must contain at least one vIOM and at least two for redundancy. As the number of vIOMs increases, MCC increases its bandwidth-processing capability.

Internal Networking Topology

The vIOM is the only VM type that terminates external-facing IP interfaces (that is, vNIC with untagged or VLAN-tagged Ethernet interfaces). Redundancy of external-facing vNICs occurs at layer 3 (that is, route resiliency with or without ECMP). BFD may be utilized to assist in liveness detection of any layer 3 interface.

The vIOM must be configured with a minimum of one and a maximum of two Base Interconnect vNICs. Heartbeats are performed over the Base Interconnect(s) with automatic failover, when appropriate.

The vIOM must be configured with a minimum of one and a maximum of two East/West Interconnect vNICs. Heartbeats are performed over the East/West Interconnect(s) with automatic failover, when appropriate.

Dynamic Scaling Considerations

The vIOM is the only VM type, other than the vMCM, that contributes per-instance configuration to the MCC. Specifically, each vIOM supports additional vNICs and therefore logical IP interfaces in the cluster. Each IP interface requires an IP address and typically requires routing level configuration (for example, BGP neighbor relationships or OSPF interfaces). Therefore, dynamic scaling requires the vIOMs to be pre-configured, including all application-level configuration associated with the vIOM. The pre-configured vIOMs may be dynamically powered up or down.

vWSM

The vWSM (*virtual Workflow Services Module*) performs all user-plane processing except that which requires TCP termination (performed on the vASM). This includes any access-side tunnel termination such as GTP-U and L2oGRE as well as any core-side termination such as GRE or L2TP/PPP. Additionally, any per-subscriber IPSEC tunnel termination (such as ePDG) is performed on the vWSM. Site-to-site IPSEC (for example, enterprise backhaul) is terminated on the vIOM. Non-TCP-termination service functions such as, PCEF, DPI, NAT, Subscriber-Firewall, and Policy-based IP steering, are performed on the vWSM.

The vWSM is DPDK-based in all deployment scenarios and therefore appears to be always running on most of its vCPUs. Application level KPIs provide the actual utilization of the vWSM (that is, idle time versus busy time).

A cluster must include at least one vWSM and two are required for redundancy. As the number of vWSMs increases, the cluster increases both the number of sessions it can terminate and the total bandwidth of all sessions.

Internal Networking Topology

The vWSM must be configured with a minimum of one and a maximum of two Base Interconnect vNICs. Heartbeats are performed over the Base Interconnect(s) with automatic failover, when appropriate.

The vWSM must also be configured with a minimum of one and a maximum of two East/West Interconnect vNICs. Heartbeats are performed over the East/West Interconnect(s) with automatic failover, when appropriate.

Dynamic Scaling Considerations

The vWSM has no per-instance configuration to which it is associated. Instead, the vWSMs are considered as a computational pool to which new sessions are assigned. As such, the vWSM may be dynamically scaled via policy-driven KPI/KCI analysis by the Acuitas VNF Manager.

Redundancy

The vWSM handles subscriber user plane processing, including certain value added services like subscriber-firewall and carrier-grade-NAT. The vWSM divides all of the sessions for which it is responsible into bundles. Each bundle of sessions is backed up to another vWSM. If a vWSM fails, all of its paired backup bundlers are activating, thus spreading the increased workload across the surviving vWSMs. The failed vWSM is also removed from the internal load balancing of new sessions until it is restored.

vSSM

The vSSM (*virtual Subscriber Services Module*) is a user-plane VM used in the compact and advanced building block modes. It is responsible for receiving packets into and out of the MMC and providing workflow services. The SSM logically combines the vIOM and vWSM capabilities, creating a single VM type that provides all user-plane services except for content services Value Added Services.

SSM VMs have external network connectivity that use SR-IOV networks (NS1, NS2). Each NS network can be configured as a flat network or with a unique VLAN ID. SSM VMs can accommodate up to 8 NS networks per SSM. A minimum of one vSSM is required and two for redundancy.

The *Integrated Services Module* (ISM) is a personality of the SSM node type. The ISM supports content and subscriber services and provides a one-stop processing node for data forwarding services and content services. The ISM is part of the integrated-services Strategy building block.

Traffic Anchoring Mode (TAM)

Traffic Anchoring Mode (TAM) is a feature supported on existing VM types, specifically the Subscriber Services Module (SSM) and Integrated Services Module (ISM). This feature applies only to SSM/ISM VM personalities where I/O and workflow services are co-located. Traffic Anchoring Mode (TAM) reduces the number of East/West hops for all user traffic. Under normal operating conditions, when combining TAM with SSM/ISM, East/West user traffic is minimized between SSMs/ISMs. TAM can operate in both MCC CUPS and non-CUPS deployments. For more information, see the *Affirmed Networks Gateway Administration Guide*.

vASM

The vASM (*virtual Advanced Services Module*) performs all TCP-termination services required in the user-plane, including TCP proxy, HTTP proxy, HTTP caching, HTTP header enrichment, HTTP video optimization, and so on.

The vASM architecture includes a small DPDK-based accelerator in all deployment scenarios and therefore appears to be always running on a subset of its vCPUs. Application level KPIs provide the actual utilization of the vASMs DPDK cores (that is, idle time versus busy time).

vASMs are not required within the cluster, however if TCP-termination user-plane services are configured, at least one vASM is required and at least two vASMs are required for redundancy. As the number of vASMs increases, the MCC cluster increases the total capacity for TCP-termination user-plane services.

Internal Networking Topology

The vASM must be configured with a minimum of one and a maximum of two Base Interconnect vNICs. Heartbeats are performed over the Base Interconnect(s) with automatic failover, when appropriate.

The vASM must also be configured with a minimum of one and a maximum of two East/West Interconnect vNICs. Heartbeats are performed over the East/West Interconnect(s) with automatic failover, when appropriate.

Dynamic Scaling Considerations

The vASM has no per-instance configuration to which it is associated. Instead, the vASMs are considered as a computational pool to which new TCP connections may be assigned. As such, the vASM may be dynamically scaled via policy-driven KPI/KCI analysis by the Acuitas VNF Manager.

Redundancy

The vASM handles advanced value added services for the subscriber user plane, including TCP proxy, HTTP proxy, content filtering, header enrichment, video optimization, and so on. There is no long-term state maintained on the vASM. If a vASM fails, it is removed from the internal load balancing of new TCP/HTTP/HTTPS flows until it is restored.

Control Plane VM

The control plane VMs consist of the vCSM, vCPM, vCCM and vDCM.

vCSM

The vCSM (*virtual Control/Content Services Module*) is a mix of user-plane and control-plane in the compact building-block mode and has a control-plane personality in the advanced building block mode. The CSM logically combines all of the vCCM, vDCM, and vASM capabilities creating a single VM type that provides all control plane plus content services Value Added Services in the user-plane.

The CSM runs the tasks needed for call control and advanced services, such as HTTP Proxy, Cache, WAP gateway, and so on. For compact deployments, the vCCM and vDCM are collapsed in the vCSM. A minimum of one vCSM is required and two for redundancy.

vCPM

The vCPM (*virtual Control Plane Module*) is a control-plane personality in the advanced building block mode. It logically combines all of the vCCM and vDCM capabilities, creating a single VM type that provides all control-plane services. The vCPM is similar to the vCSM, except that it does not provide the content services Value Added Services in the user-plane. A minimum of one vCPM is required and two for redundancy.

vCCM

The vCCM (*virtual Centralized Control Module*) performs all cluster-level control plane functions, including IP routing (e.g., BGP, OSPF) and certain centralized control plane coordination functions (for example, session-level load balancing to vDCM).

A cluster must contain at least one vCCM and two are required for redundancy. There is no limit to the number of vCCMs that can be deployed, however most clusters are adequately supported by a maximum of two vCCMs.

Internal Networking Topology

The vCCM must be configured with a minimum of one and a maximum of two Base Interconnect vNICs. Heartbeats are performed over the Base Interconnect(s) with automatic failover, when appropriate.

The vCCM must also be configured minimum of one and a maximum of two East/West Interconnect vNICs. Heartbeats are performed over the East/West Interconnect(s) with automatic failover, when appropriate.

Dynamic Scaling Considerations

The vCCM has no per-instance configuration to which it is associated. Instead, vCCMs are considered as a computational pool to which centralized control plane processes may be assigned. While the vCCM architecturally supports dynamic scaling, there is typically no need to include this VM type in the dynamic scaling policy.

Redundancy

The vCCM handles centralized control plane protocols such as OSPF and BGP, as well as certain internal coordinating functions, particularly related to internal load balancing decisions (for example, assigning new PDN sessions to vDCMs). Although more than two vCCMs are supported in a cluster, MCC requires at least two vCCMs for resiliency. Each process that maintains important in-memory state is paired with a similar process running on another vCCM. If a vCCM fails, all of its processes are failed over to their pre-instantiated backups. Similarly, if a single process fails on the vCCM, it is individually failed over.

vDCM

The vDCM (*virtual Distributed Control Module*) performs all internally scalable distributed control plane duties within the MCC cluster. This primarily includes the multi-access call-control (that is, *gateway* functionality). Just as the vWSM serves as the user-plane anchor for individual sessions, the vDCM serves as the control plane anchor.

A cluster must contain at least one vDCM and two are required for redundancy. As the number of vDCMs increases, the MCC cluster increases the number of sessions it can terminate.

Internal Networking Topology

The vDCM must be configured with a minimum of one and maximum of two Base Interconnect vNICs. Heartbeats are performed over the Base Interconnect(s) with automatic failover, when appropriate.

The vDCM must also be configured with a minimum of one and maximum of two East/West Interconnect vNICs. Heartbeats are performed over the East/West Interconnect(s) with automatic failover, when appropriate.

Dynamic Scaling Considerations

The vDCM has no per-instance configuration to which it is associated. Instead, vDCMs are considered as a computational pool to which new sessions may be assigned. As such, the vDCM may be dynamically scaled via policy-driven KPI/KCI analysis by the Acuitas VNF Manager.

Redundancy

The vDCM handles all call control related duties including GTP-C, RADIUS, and Diameter protocol termination. Each vDCM instantiates multiple call-control processes. Each call control process is, in turn, associated with a backup on another vDCM. When a vDCM fails, all of its paired backup call control processes are activated, thereby spreading the increased workload across the surviving vDCMs. If a single call control process fails on the vDCM, it is individually failed over. The failed vDCM is also removed from the internal load balancing of new sessions until it is restored.

Configurable Base IP Addresses on VM Interfaces

Each VM has interfaces that require a certain number of IP addresses, including:

- MCM — Requires an IP address for EMS connectivity and a base IP address for control-plane communication between VMs in the cluster.
- SSM — Requires a base IP address for control-plane communication between VMs in the cluster, a data fabric address (optional) for data plane communication between CSMs/SSMs in the cluster, and some number of external IP addresses for the traffic that enters/exits the cluster.
- CSM — Requires a base IP address for control-plane communication between the VMs in the cluster and a data fabric address (optional) for data plane communication between the CSMs/SSMs in the cluster.

Operators can override the default base address and manually configure IPv4/IPv6 addresses on the VM base interfaces. The default base addresses are in the form of `169.254.cid.nid` where:

- `cid` is the cluster ID

- `nid` is the node ID

Using this feature, the Operator selects a different base address for each node. However, the base addresses across all nodes within the cluster must belong to a common subnet.

***Note:** In redundant MCC systems, Base and Base 2 Ethernet ports are no longer bonded. Each separate base port is assigned a unique MAC address.*

L3 IPv4/IPv6 Data Fabric Connectivity

In virtualized deployments, the fabric network is the customer's data-center network. Since proprietary ether types are not allowed, this feature allows for an IPv4/IPv6 encapsulation of data plane packets. When selecting 'L3' as the data plane protocol (default is L2), a configured data fabric IP address for each SSM/CSM node is required, and these addresses must belong to a common IP subnet. Note that in either mode (L2 or L3), an optional VLAN tag can be configured for the data fabric ports.

The entire cluster must use either L3 (Layer-3) or L2 (Layer-2) mode. You configure the IP address under `cluster <name> node <name>` and configure the protocol type for Path-A and Path-B under `cluster <id> data-plane-protection`.

MCC supports both L2 East/West (EW) and L3 EW. When you upgrade a cluster to a release supporting L2/L3 EW, the cluster automatically defaults to the existing L2 solution. When operating in L2 mode, existing cluster-level parameters are used.

L3 EW supports fabric redundancy and each node has two fabric interfaces, each configured for different L3 subnets. MCC continually monitors fabric connectivity after the node is fully operational. MCC removes a node from service if fabric connectivity is lost or degraded.

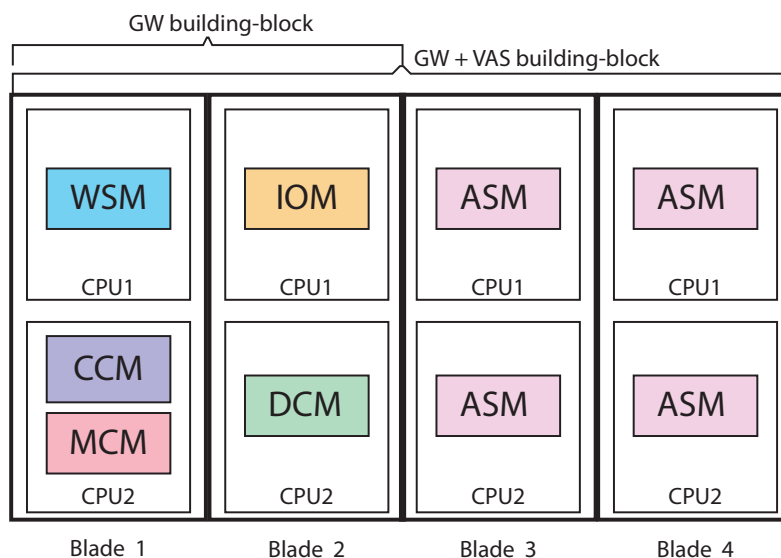
Gateway Building Block

A gateway building block is made up of two server building blocks. At least two building blocks are required in a cluster to support N-way redundancy. Additional building blocks can be added to reach the required capacity and the maximum number of supported VMs. To support *both gateway and content services*, a building block requires the following VMs:

- One vIOM
- One vWSM
- One vMCM – Two building blocks in the cluster must contain the vMCM.
- One vDCM – Two are required in a cluster.

- One vCCM
- Four vASMs (for content services support)

Figure 4-3 MCC VM Building Blocks



Additional VM Types/Deployment Models

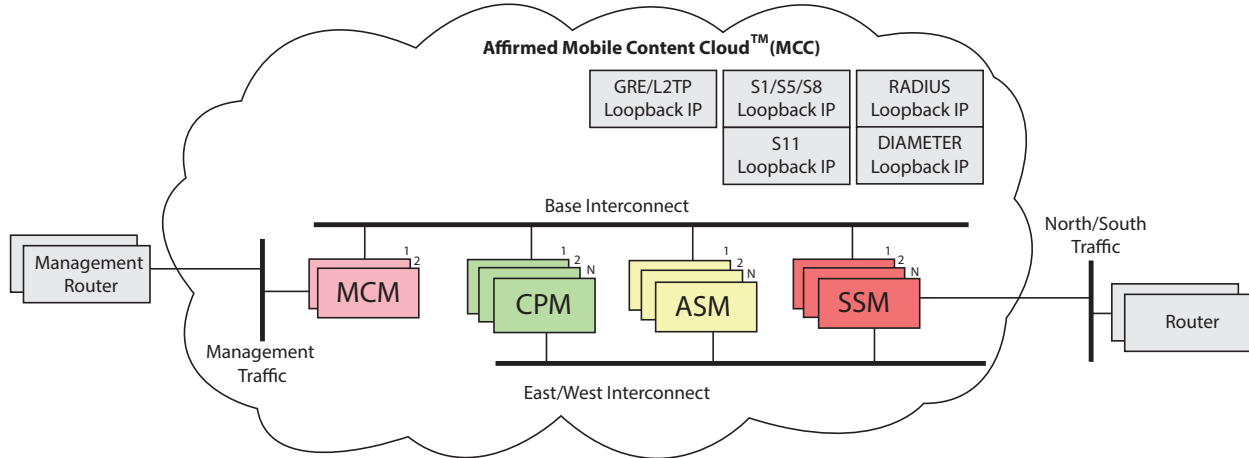
In order to achieve a smaller overall footprint when dynamic scale is not required, Affirmed Networks supports various VM combination types to facilitate a smaller overall VM count within the cluster. This section describes the following additional deployment models:

- [Compact Deployment Model \(with Content Services\)](#)
- [Advanced-Services-Separation Deployment Model \(without Content Services\)](#)

Compact Deployment Model (with Content Services)

A minimum of three VM types is required to support full cluster capability, including content services.

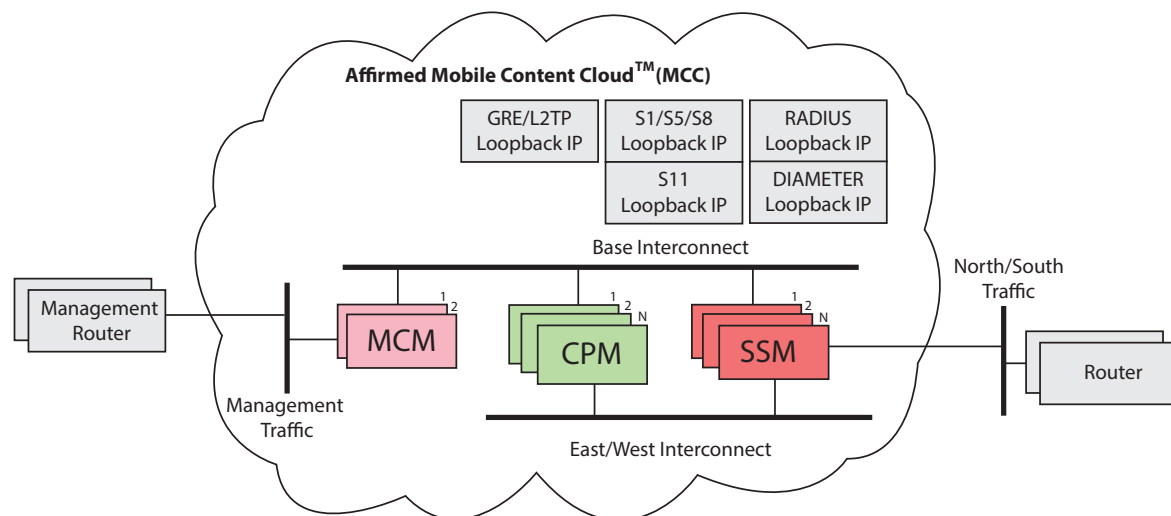
- vMCM
- vCSM
- vSSM

Figure 4-4 Advanced-Svcs-Separation Deployment (with Content Services)

Advanced-Services-Separation Deployment Model (without Content Services)

This cluster style is an adaptation of the [Compact Deployment Model \(with Content Services\)](#) for cases where content services in the user-plane is not required. The CSM in the former model is replaced with the CPM.

- vMCM
- vCPM
- vSSM

Figure 4-5 Small Deployment Model (without Content Services)

Capacity and Performance

The capacity and performance of an MCC cluster depends on the number and type of VMs initialized in the cluster. Since the MCC supports many combinations of different VM types in a cluster, it may be useful to create templates or building blocks to build a cluster configuration to meet certain capacity requirements. The following section describes two different types of building blocks; gateway only services and both gateway and content services.

***Note:** These example building block scenarios show how to build capacity using various VM types. Depending on the particular call model requirements, your cluster design may differ from the example.*

Gateway Services Building Block

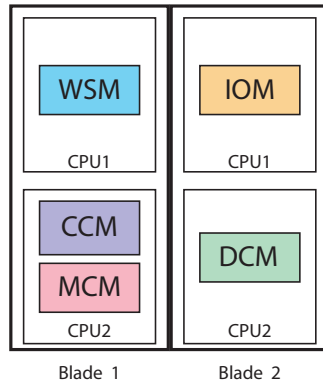
To provide *gateway services only*, a building block requires the following VMs:

- One vIOM
- One vWSM
- One vMCM – Two building blocks in the cluster must contain the vMCM.
- One vDCM – Two are required in a cluster.
- One vCCM

Figure 4-6 shows an example gateway services building block. At least two building blocks are required in a cluster to support N-way redundancy. In Figure 4-6, the vIOM and the vWSM are anchored to CPU1 on blade 1 and blade 2. A vDCM is installed in CPU2 in either of the blades. The vMCM and vCCM share CPU2.

The following guidelines apply to a gateway services building block:

- A cluster must contain at least two building blocks to ensure VM redundancy
- More than two vMCMs are not required in a cluster that contains more than two building blocks
- A cluster must contain at least two vCCMs for redundancy. Depending on the call model, more than two vCCMs may be required.

Figure 4-6 Gateway Services Building Block

Example Configuration

This section describes the capacity and throughput for an example configuration. The capacity and throughput for your specific configuration may differ.

In an HP C7000 chassis with 10 Gb switch fabrics, one building block supports:

- 1 million sessions (1M default + 1M dedicated bearers)
- 10 Gbps of HTTP/TCP traffic (with average downlink packet size of 1060 bytes and average uplink packet size of 70 bytes)
- A DPI HTTP/TCP throughput of 10 Gbps (CPU capacity exists too support the same throughput even with DPI enabled)
- A session setup rate of ~ 5,000 per second

Figure 4-7 shows an example of a cluster with a loaded HP C7000 chassis. In this example, eight building blocks are used to fill all 16 available blades in the chassis.

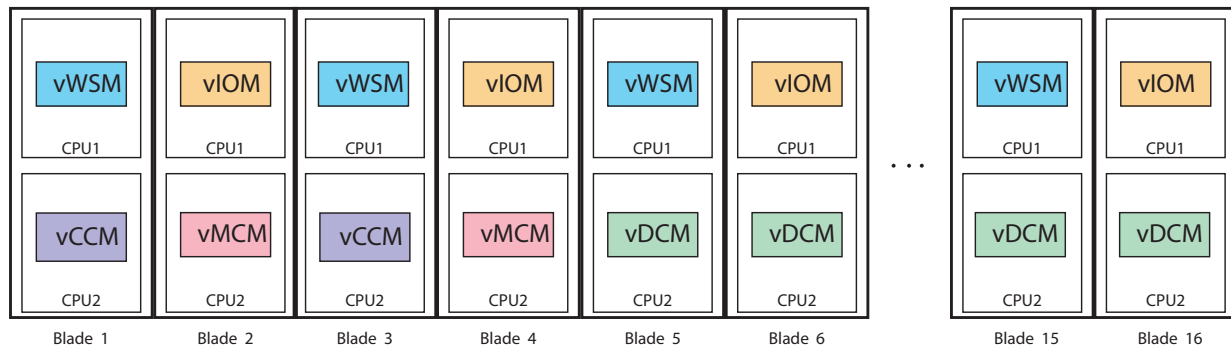
The first two building blocks contain two vCCMs and two vMCMs to serve the entire cluster. In the remaining building blocks, vDCMs occupy the second CPUs in the blades. One of the eight building blocks provides throughput redundancy.

In this example configuration, a loaded HP C7000 supports:

- 7 million sessions (7M default + 7M dedicated bearers)
- 70 Gbps of HTTP/TCP traffic (with average downlink packet size of 1060 bytes and average uplink packet size of 70 bytes)
- A DPI HTTP/TCP throughput of 70 Gbps (there is CPU capacity to support the same throughput even with DPI enabled)
- A session setup rate of ~35,000 per second

A cluster is not confined to a single HP C7000 enclosure. The example in [Figure 4-7](#) provides a simple view to compare capacity within a traditional ATCA solution. Operators can also configure a cluster using building blocks in different HP C7000 chassis. Provided that adequate external switching bandwidth exists for east-west and control traffic, a cluster can span multiple chassis, thereby reducing the impact of a chassis failure.

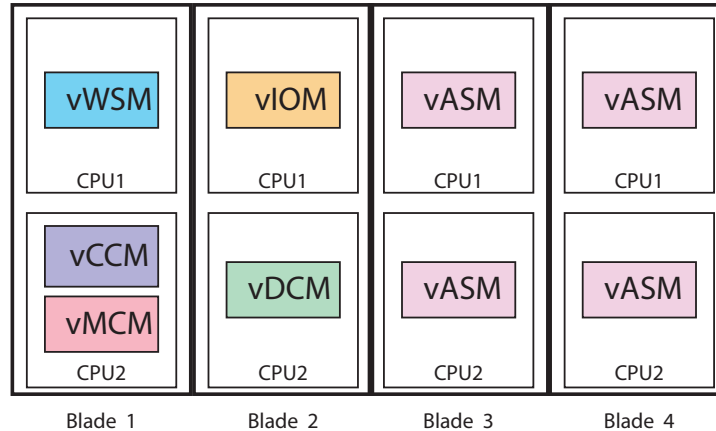
Figure 4-7 HP C7000 Gateway Cluster Example



Gateway and Content Services Building Block

To support both gateway and content services, a building block must consist of the following VMs:

- One vIOM
- One vWSM
- One vMCM – Two building blocks in the cluster must contain the vMCM.
- One vDCM – Two are required in a cluster.
- One vCCM
- Four vASMs

Figure 4-8 Gateway and Content Services Building Block

In the gateway and content services building block, four ASMs are required to support the content services. This configuration requires a minimum of four blades for optimal performance. [Figure 4-8](#) shows one possible example of this building block to optimize performance. Other configurations are also supported.

In the example in [Figure 4-8](#), blade 1 and blade 2 are the same as [Gateway Services Building Block](#). Blade 3 and blade 4 contain four vASMs responsible for content services.

The following guidelines apply to the gateway and content services building block:

- A cluster must contain at least two building blocks to ensure VM redundancy
- More than two vMCMs are not required in a cluster that contains more than two building blocks
- Two vCCMs within the cluster are required for redundancy. Depending on the call model, additional vCCMs may be required
- Each VM uses 64 GB of memory on each blade.
- The four vASMs provide enough processing capacity to match the 10Gbps throughput supported by the gateway building block (the first 2 blades). Each blade containing two ASMs supports 3.75 Gbps of proxy throughput.

Example Configuration

This section describes the capacity and throughput for an example configuration. The capacity and throughput for your specific configuration may differ.

In an HP C7000 chassis with 10 Gb switch fabrics, one building block supports:

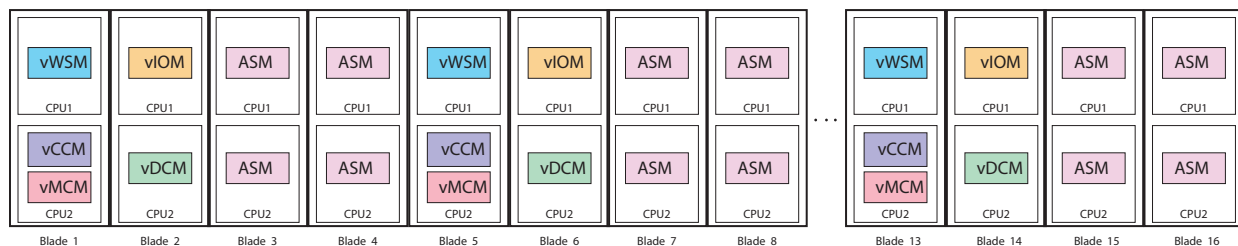
- 1 million sessions (1M default + 1M dedicated bearers)
- 10 Gbps of HTTP/TCP traffic (with average downlink packet size of 1060 bytes and average uplink packet size of 70 bytes)
- A DPI HTTP/TCP throughput of 10 Gbps (CPU capacity exists too support the same throughput even with DPI enabled)
- A session setup rate of ~ 5,000 per second
- A proxy throughput of ~ 7.5 Gbps. If other optimization services are enabled (that is, video optimization) the two blades allocated for vASMs may not be able to support the required proxy throughput as CPU resources are used to support the optimization services. In this case, increase the number of blades containing the vASMs.

Figure 4-9 shows an example of a cluster with a loaded HP C7000 chassis. This example uses four gateway and content building blocks to fill all 16 available blades in the chassis.

In this example configuration, a loaded HP C7000 supports:

- 3 million sessions (7M default + 7M dedicated bearers)
- 30 Gbps of HTTP/TCP traffic (with average downlink packet size of 1060 bytes and average uplink packet size of 70 bytes)
- A DPI HTTP/TCP throughput of 30 Gbps (there is CPU capacity to support the same throughput even with DPI enabled)
- A session setup rate of ~15,000 per second
- A proxy throughput of 30 Gbps

Figure 4-9 HP C7000 Gateway and Services Cluster Example



VM Cluster Optimization

You can configure performance optimization for SSM, IOM, WSM, ISM and ASM VMs based on the node's projected traffic throughput. Typically configured during an MCC deployment, this feature provides CLI commands to configure the number of cores that will be used for I/O on an SSM/IOM/WSM/ISM VM. These configurations are at a cluster level and apply to all the SSM/IOM/WSM/ASM VMs in the cluster.

When the I/O core level functionality is configured using these commands, a warning message will prompt the user to proceed, noting that all changes will be applied to the cluster only after the affected nodes in the cluster are rebooted.

For information about configuring the I/O core level, see the applicable deployment guides.

Statement of Compliance

[Table A-1](#) lists the MCC Statement of Compliance (SoC) with Technical Specifications (TS), standards, and Request for Comments (RFC).

Table A-1 MCC Statement of Compliance

Feature/Function	Technical Specifications, Standards, and Request for Comments
BFD	■ RFC 5880: Bidirectional Forwarding Detection (BFD) ■ RFC 5881: Bidirectional Forwarding Detection (BFD) for IPv4 and IPv6 (Single Hop) ■ RFC 5883: Bidirectional Forwarding Detection (BFD) for Multihop Paths
Bp	■ 3GPP TS 32.251: Telecommunications management; Charging management; Packet Switched (PS) domain charging, Revision 9.9 ■ 3GPP TS 32.297: Telecommunications management; Charging management; Charging Data Record (CDR) file format and transfer, Revision 9 ■ 3GPP TS 32.298: Telecommunications management; Charging management; Charging Data Record (CDR) parameter description, Revision 9
Ga	■ 3GPP TS 32.251: Telecommunications management; Charging management; Packet Switched (PS) domain charging, Revision 9.9 ■ 3GPP TS 32.295: Telecommunications management; Charging management; Charging Data Record (CDR) transfer, Revision 9 ■ 3GPP TS 32.298: Telecommunications management; Charging management; Charging Data Record (CDR) parameter description, Revision 9
Gi/Sgi	■ 3GPP TS 29.061: Interworking between the Public Land Mobile Network (PLMN) supporting packet based services and Packet Data Networks (PDN), Revision 9 ■ RFC 2865: Remote Authentication Dial In User Service (RADIUS) ■ RFC 2866: RADIUS Accounting ■ RFC 3576: Dynamic Authorization Extensions to Remote Authentication Dial In User Service (RADIUS) ■ RFC 4818: RADIUS Delegated-IPv6-Prefix Attribute ■ RFC 3162: RADIUS and IPv6 ■ RFC 2869: RADIUS Extensions ■ RFC 2868: RADIUS Tunnel Authentication Attributes

Table A-1 MCC Statement of Compliance (Continued)

Feature/Function	Technical Specifications, Standards, and Request for Comments
Gn/Gp	■ 3GPP TS 29.060 V9.3.0 (2010-06): General Packet Radio Service (GPRS);GPRS Tunneling Protocol (GTP) across the Gn and Gp interface ■ 3GPP TS 29.274 V9.3.0 (2010-06): 3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunneling Protocol for Control plane (GTPv2-C); Stage 3 ■ 3GPP TS 23.401 V9.3.0 (2009-12): General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access ■ 3GPP TS 09.60 V7.10.0 Release 1998: Digital cellular telecommunications system (Phase 2+); General Packet Radio Service (GPRS); GPRS Tunneling Protocol (GTP) across the Gn and Gp Interface
Gx	■ 3GPP TS 23.003 V12.0.0 (2013-09): Technical Specification Group Core Network and Terminals; Numbering, addressing and identification ■ 3GPP TS 29.212 V12.7.0 (2014-12): Technical Specification Group Core Network and Terminals; Policy and Charging Control (PCC); Reference points ■ 3GPP TS 29.214: Technical Specification Group Core Network and Terminals; Policy and Charging Control over Rx reference point (Release 13) ■ 3GPP TS 32.299 V10.7.0 (2012-09): Telecommunication management; Charging management; Diameter charging application ■ 3GPP TS 29.229 V10.0.0 (2011-03): Cx and Dx interfaces based on the Diameter protocol; Protocol detail ■ 3GPP TS 29.214 V10.0.0 (2010-09): Policy and Charging Control over Rx reference point ■ RFC 3588: Diameter Based Protocol ■ RFC 3539: Authentication, Authorization and Accounting (AAA) Transport Profile ■ RFC 4006: Diameter Credit-Control Application ■ RFC 4282: The Network Access Identifier ■ RFC 6733: Diameter Base Protocol
Gy	■ 3GPP TS 32.299 V10.7.0 (2012-09): Telecommunication management; Charging management; Diameter charging application ■ 3GPP TS 32.299 V15.3.0 (2018-06): Telecommunication management; Charging management; Diameter charging application <i>Note: Partial compliance. Compliance with 5G-NSA attributes in V15.3.0.</i> ■ 3GPP TS 32.251 V9.9.0 (2012-06): Telecommunication management; Charging management; Packet Switched (PS) domain charging ■ RFC 3588: Diameter Based Protocol ■ RFC 3539: Authentication, Authorization and Accounting (AAA) Transport Profile ■ RFC 4006: Diameter Credit-Control Application ■ RFC 6733: Diameter Base Protocol
Gz	■ 3GPP TS 32.251: Telecommunications management; Charging management; Packet Switched (PS) domain charging, Revision 9.9 ■ 3GPP TS 32.299: Telecommunications management; Charging management; Diameter charging applications, Revision 10.7 ■ RFC 3588: Diameter Base Protocol
Internet Protocol	■ RFC 791: Internet Protocol

Table A-1 MCC Statement of Compliance (Continued)

Feature/Function	Technical Specifications, Standards, and Request for Comments
IPSec Protocols	■ 3GPP TS 33.210: 3G security; Network Domain Security (NDS); IP network layer security, Revision 13 ■ RFC 2409: The Internet Key Exchange (IKE) ■ RFC 5996: Internet Key Exchange Protocol Version 2 (IKEv2) ■ RFC 4303: IP Encapsulating Security Payload (ESP) ■ RFC 3947: Negotiation of NAT-Traversal in the IKE
Lawful Interception	■ 3GPP TS 33.106 V15.1.0: 3G security; Lawful Interception Requirements (Release 15) ■ 3GPP TS 33.107 V15.4.0: 3G security; Lawful Interception Architecture and Functions (Release 15) ■ 3GPP TS 33.108 V15.3.0: 3G security; Handover Interface for Lawful Interception (LI) (Release 15) ■ RFC 2126: ISO Transport Service on Top of TCP (ITOT) ■ ETSI TS 101 671 V3.14.1: Lawful Interception (LI); Handover Interface for the Lawful Interception of Telecommunications Traffic
Network Management	■ RFC 1213: Management Information Base for Network Management of TCP/IP-based internets: MIB-II ■ RFC 1607: A Historical Perspective on the Usage of IP Version 9 ■ RFC 1901: Introduction to Community-based SNMPv2 ■ RFC 2578: Structure of Management Information Version 2 (SMIv2) ■ RFC 2579: Textual Conventions for SMIv2 ■ RFC 2580: Conformance Statements for SMIv2 ■ RFC 3416: Version 2 of the Protocol Operations for the Simple Network Management Protocol (SNMP) ■ RFC 3418: Management Information Base (MIB) for the Simple Network Management Protocol (SNMP) ■ RFC 4741: NETCONF Configuration Protocol ■ RFC 4742: Using the NETCONF Configuration Protocol over Secure Shell (SSH) ■ RFC 5717: Partial Lock Remote Procedure Call (RPC) for NETCONF ■ RFC 5277: NETCONF Event Notifications ■ RFC 6020: YANG - A Data Modeling Language for the Network Configuration Protocol (NETCONF) ■ RFC 6021: Common Yang Data Types ■ RFC 6022: YANG Module for NETCONF Monitoring ■ RFC 6241: Network Configuration Protocol (NETCONF) ■ RFC 6242: Using the NETCONF Protocol over Secure Shell (SSH) ■ RFC 6243: With-defaults Capability for NETCONF
openssh RFC	■ Other than Protocol 1 (which is disabled), the following are used directly: http://www.openssh.com/specs.html

Table A-1 MCC Statement of Compliance (Continued)

Feature/Function	Technical Specifications, Standards, and Request for Comments
OCS/OFCS	■ 3GPP TS 32.240: Telecommunications management; Charging management; Charging architecture and principles, Revision 9 ■ 3GPP TS 32.299: Telecommunications management; Charging management; Diameter charging applications, Revision 10.7
PCEF	■ 3GPP TS 23.207: End-to-end Quality of Service (QoS) concept and architecture, Revision 9 ■ 3GPP TS 29.213 V11.5.0 (December 2012): Policy and charging control signaling flows and Quality of Service (QoS) parameter mapping
S1-U	■ 3GPP TS 29.281: GPRS Tunneling Protocol User Plane (GTPv1-U), Revision 9.3 ■ 3GPP TS 36.414: Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 Data Transport, Revision 9
Routing Protocols	■ RFC 2439: BGP Route Flap Damping ■ RFC 2545: Use of BGP-4 Multiprotocol Extensions for IPv6 Inter-Domain Routing ■ RFC 1765: OSPF Database Overflow ■ RFC 2328: OSPF Version 2 ■ RFC 2740: OSPF for IPv6 ■ RFC 2858: Multiprotocol Extensions for BGP-4 ■ RFC 3101: The OSPF Not-So-Stubby Area (NSSA) Option ■ RFC 3107: Carrying Label Information in BGP-4 ■ RFC 3623: Graceful OSPF Restart ■ RFC 4364: BGP/MPLS VPNs ■ RFC 4760: Multiprotocol Extensions ■ RFC 4893: BGP Support for Four-Octet AS Number Space ■ RFC 5187: OSPFv3 Graceful Restart ■ RFC 5243: OSPF Database Exchange Summary List Optimization (OSPFv2 and OSPFv3)
S4, S5, S8, S11	■ 3GPP TS 23.401: General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access, Revision 9.3 ■ 3GPP TS 29.274: Evolved GPRS Tunneling Protocol for Control plane (GTPv2-C), Revision 9.3
S2a, S2b	■ 3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunneling Protocol for Control Plane (GTPv2-C); Stage 3 ■ 3GPP TS 23.401 V9.3.0 (2009-12): General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access ■ 3GPP TS 29.281 V9.3.0 (2010-06): General Packet Radio System (GPRS) Tunneling Protocol User Plane (GTPv1-U) (Release 9) ■ 3GPP TR 23.852 V12.0.0 (2013-09): Study on S2a Mobility based on GPRS Tunneling Protocol (GTP) and Wireless Local Area Network (WLAN) access to the Enhanced Packet Core (EPC) network (SaMOG); Stage 2 ■ 3GPP TS 23.402 V12.0.0 (2013-03): Technical Specification Group Services and System Aspects; Architecture Enhancements for non-3GPP Access ■ 3GPP TR 23.834: GPRS Tunneling Protocol (GTP) based S2b

Table A-1 MCC Statement of Compliance (Continued)

Feature/Function	Technical Specifications, Standards, and Request for Comments
S6b, SWm	■ 3GPP TS 29.273, V12.5.0 Release 12; UMTS/ LTE Evolved Packet System (EPS) 3GPP EPS AAA Interfaces ■ 3GPP TS 23.402, V10.4.0 Release 10; UMTS/LTE, Architecture Enhancements for non-3GPP Access
Sxa, Sxb	■ 3GPP TS 23.214 V15.2.0 (2018-03): Architecture enhancements for control and user plane separation of EPC nodes ■ 3GPP TS 29.244 V14.4.0 (2018-06): Interface between the Control Plane and the User Plane of EPC Nodes
Security and Encryption	■ ISO 17799:2005:Information technology - Security techniques - Code of practice for information security management ■ ISO 27001:Information technology - Security techniques - Information security management systems -Requirements ■ 3GPP TS 33.102:3G security; Security architecture ■ 3GPP TS 33.210: Network Domain Security (NDS); Authentication Framework (AF) ■ RFC 2246: The TLS Protocol ■ RFC 4251: The Secure Shell (SSH) Protocol Architecture ■ RFC 4252: The Secure Shell (SSH) Authentication Protocol ■ RFC 4254: The Secure Shell (SSH) Connection Protocol ■ RFC 4346: The Transport Layer Security (TLS) Protocol ■ RFC 7296: Internet Key Exchange Protocol Version 2 (IKEv2) ■ RFC 3602: The AES-CBC Cipher Algorithm and Its Use with IPsec ■ RFC 4868: Using HMAC-SHA-256, HMAC-SHA-384, and HMAC-SHA-512 with IPsec
Tunneling	■ IEEE 802.1Q:IEEE Standard for Local and metropolitan area networks - Media Access Control (MAC) Bridges and Virtual Bridged Local Area Networks ■ 3GPP 29.281: General Packet Radio System (GPRS) Tunneling Protocol User Plane (GTPv1-U), Release 9.3 ■ 3GPP TS 29.274 V12.8.0 (2014-03): 3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunneling Protocol for Control plane (GTPv2-C); Stage 3 ■ RFC 1662: PPP in HDLC-like Framing ■ RFC 2784: Generic Routing Encapsulation (GRE)

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